The Holonet Machine

A complete engineering blueprint for a computer
whose wiring diagram is a piece of finite geometry

Every promoted figure is typed: proved, measured, published, modelled, or open.

W(3,3)–E_8 project, glue track

3 August 2026 · Passes 2700–2907

Abstract

This is a build document for a computer that does not exist yet.

Most computer designs begin with a goal and invent a structure to reach it. This one runs the other way. It starts from a piece of pure mathematics — a forty-point shape called $W(3,3)$, studied since the 1900s for reasons having nothing to do with computing — and finds that the shape already contains an instruction set, a memory layout, a network topology, an error model, and a set of quantum resources. Nothing here was designed in the usual sense. It was read off.

The result is unusually specific for something unbuilt. The instruction set needs *four* operations in two bits, and they generate a group of exactly 4,199,040 transformations, with a worst-case program length of exactly **19**. On the HX8K target the minimal arithmetic core uses **43 logic cells** at **208.86 MHz**; on the UP5K it uses the same 43 cells at 72.40 MHz. Those are part-specific measurements, not interchangeable headline numbers. A support-only observation discards exactly 8/3 bits of conditional phase information. Landauer supplies a thermodynamic floor for erasing that information, not an engineering power forecast. Published time-bin/frequency-bin experiments report a two-qutrit gate fidelity near 0.92; this is prior-art evidence for a component class, not a measurement of the unbuilt Holonet stack. And thirty-six of the forty points turn out to be exactly the machine's magic states, so the quantum resource is not an add-on — it is everything except one line of the address space.

We give the full instruction semantics, the datapath, the measured cell budget, the network protocol, the thermodynamic floor, and the verification methodology. We also give — because this is what makes a blueprint trustworthy rather than merely confident — an itemised record of every figure we previously got wrong, how each error survived exhaustive testing, and what finally caught it.

Contents

1 How to read this document

This document has two readers in mind and does not compromise for either.

If you are not an engineer.
Read the cream boxes and the diagrams. They are complete: every idea in the machine is explained there from scratch, in order, with no symbols you have not already met. You may skip every blue box without losing the thread.

If you are an engineer.
The blue boxes carry the specification: exact semantics, matrices, cell counts, clock frequencies, proof obligations and the commands that reproduce them. The cream boxes are there so you can hand this to a colleague in a different field.

Three habits of this document, stated up front so they are not mistaken for style:

1. **Every number is measured or proved, and says which.** There are no estimates presented as results. Where an estimate appears it is labelled *modelled* and cannot be cited as anything else.
2. **Nothing is claimed to be new without checking.** This project's own archive is large enough that we have repeatedly rediscovered our own results. Where something is someone else's, it says so.
3. **Mistakes stay in.** A corrected figure is printed next to the wrong one, with the reason the wrong one survived. §19 collects them.

That third kind of box appears throughout:

What we got wrong, and how we know. These record something this project believed, published, and then had to withdraw — with the measurement that overturned it. The errata index is authoritative and grows when a claim is corrected. These boxes are not apologies; they are the reason the other numbers in this document can be trusted. A blueprint with no errata is a blueprint nobody has checked.

2 The machine in one page

In plain language.*Every computer you have used stores information as* bits *— switches that are either off or on. This machine stores information in* qutrits*: units with three settings instead of two. That sounds like a small change. It is not, because of what the three settings let you do with* geometry*.*
Here is the strange and central fact. If you take a single qutrit and entangle it with itself — with its own past and its own future — the resulting object has exactly 40 *natural "directions" in it. Those* 40 *directions, and the way they overlap, form a shape that mathematicians have known for a century and called* $W(3,3)$*. The machine we are describing does not* use *that shape as a convenience. The shape* is *the machine: its memory layout, its instruction set, its network topology, and its error model are all readings of the same* 40*-point object.*

One qutrit, entangled with its own past, is enough to generate the entire substrate. The 40 points of $W(3,3)$ are not 40 things — they are 40 views of one thing.

The machine, specified.

Logical substrate	$W(3,3)$: the symplectic generalised quadrangle of order $(3,3)$. 40 points, 40 totally isotropic lines, collinearity graph $\mathrm{SRG}(40,12,2,4)$.
State	One qutrit pair (past, future) living in $\mathbb{C}^9 = \mathrm{End}(\mathbb{C}^3)$.
Tracked quantity	The *Pauli frame*: four trits $(x_p, z_p, x_f, z_f) \in \mathbb{F}_3^4$, i.e. 81 states.
Public ISA	$\mathcal{I}_{\mathrm{holo}}$, eight opcodes, three bits.
Internal micro-ISA	F_p, $\mathrm{CX}_{p\to f}$, $\mathrm{CX}_{f\to p}$, Z_p — four operations, two bits, no illegal encoding.
Group generated	$\mathrm{ASp}(4,3) = \mathbb{F}_3^4 \rtimes \mathrm{Sp}(4,3)$, order $81 \times 51840 = 4{,}199{,}040$ *exactly.*
Arithmetic unit	HX8K: minimal 43 LC at 208.86 MHz; public 72 LC at 175.72 MHz; minimal plus support 48 LC at 207.64 MHz. UP5K: the corresponding clocks are 72.40, 60.80, and 72.05 MHz.
Physical layer	Time-bin × frequency-bin photonic qutrits; the two-qutrit SUM gate is experimentally demonstrated at fidelity 0.92 ± 0.01.
Network	40-ary fractal; $N_n = 40^n$ leaves, routing depth $8n = 8\log_{40} N$; the routing code satisfies Kraft equality exactly, so *every* bit string is a legal address.

The whole system, at a glance

NETWORK — 40-ary fractal holonet; remote SUM costs one Bell pair + two classical trits §14

↓

INSTRUCTION SET — eight opcodes; generates $\mathrm{ASp}(4,3)$, order 4,199,040 §6

↓

FRAME UNIT — 7-bit frame code; minimal execution 43 LC; support readout adds 5 LC §7

↓

GEOMETRY — $W(3,3)$: 40 points, 40 lines, $\mathrm{SRG}(40,12,2,4)$ §4

↓

PHYSICS — one self-entangled qutrit; time bins × frequency bins §3

The unusual property of this stack is that the layers are not independent design choices stacked by convention. Each one is *forced* by the one below it. The network is 40-ary because the geometry has 40 points. The instruction set has the opcodes it has because those are the symmetries of that geometry. This is what the rest of the document demonstrates, layer by layer.

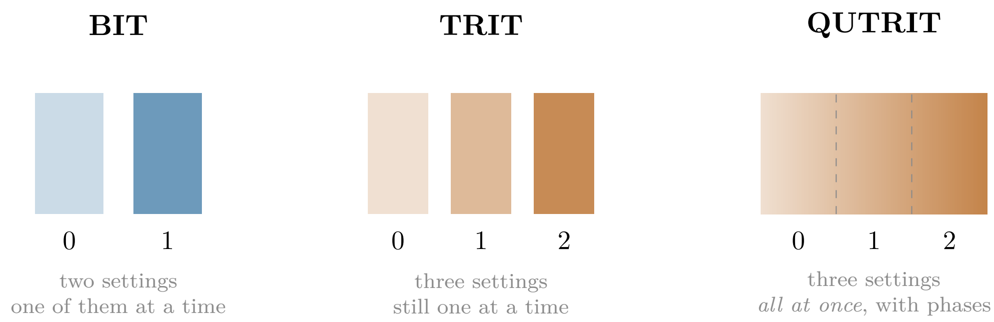
Part I — The object

One finite geometry, defined by a single congruence, and everything it already contains before anyone designs anything.

3 Qutrits, and what "self-entangled" means

3.1 From bits to qutrits

In plain language. *A bit is a switch: off or on,* 0 *or* 1*. A* trit *is a three-way switch:* 0*,* 1*, or* 2*. Ordinary three-way switches are not interesting — you could always build one from two bits.*
A qutrit *is different. It is a physical three-way switch obeying quantum mechanics, which means it can be in a blend of all three settings at once, and — this is the part that matters —* two *qutrits can be correlated in ways that no pair of classical switches can imitate.*



Exactly. A qutrit is a unit vector in $\mathbb{C}^3$. Two qutrits live in $\mathbb{C}^3 \otimes \mathbb{C}^3 = \mathbb{C}^9$. The relevant operators are generated by

$$X\,|j\rangle = |j+1 \bmod 3\rangle\,, \qquad Z\,|j\rangle = \omega^j\,|j\rangle\,, \qquad \omega = e^{2\pi i/3},$$

which satisfy $ZX = \omega XZ$. Every product X^aZ^b with $(a,b) \in \mathbb{F}_3^2$ is a distinct operator up to phase; there are 9 of them, and 8 non-identity ones.

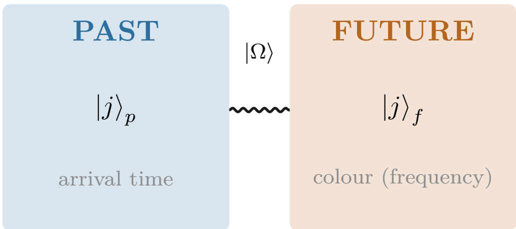
3.2 The self-entanglement, and where the 40 comes from

In plain language. *Now the key move. Take* one *qutrit and entangle it with itself at two different times — call them "past" and "future". The proposed optical encoding assigns one qutrit to arrival-time bins and one to frequency bins on a single carrier. The words "past" and "future" name these two logical registers; they do not assert a demonstrated closed-time-loop interaction or a new physical form of temporal entanglement.*
Mathematically, an operator acting on one qutrit is a 3×3 table of numbers, and there are 9 independent slots in such a table. So the space of things you can do *to one qutrit is itself 9-dimensional — exactly the size of the space of* states *of two qutrits. This is not a coincidence or an analogy. It is an identity:*

$$\underbrace{\mathbb{C}^3 \otimes \mathbb{C}^3}_{\textit{two qutrits}} = \underbrace{\mathrm{End}(\mathbb{C}^3)}_{\textit{operations on one qutrit}} .$$

So "the two-qutrit machine" and "the one-qutrit machine that can act on itself" are the same machine described twice.
Now count the directions. There are 81 ways to label a pair (four trits), one of which is "do nothing". That leaves 80. Each direction and its opposite behave identically for our purposes, and each direction comes in 2 such flavours, so the distinct directions number $80/2 = 40$.

$$\frac{3^4-1}{2} = \frac{81-1}{2} = 40.$$ **The forty points of the geometry are the forty independent things one self-entangled qutrit can be measured along.**



$$|\Omega\rangle = \mathrm{CX}_{p\to f}\,(F_3 \otimes I)\,|0\rangle_p\,|0\rangle_f = \frac{1}{\sqrt{3}} \sum_{j=0}^{2} |j\rangle_p\,|j\rangle_f$$

One optical carrier, two encoded qutrit degrees of freedom. The past/future names are architectural labels.

What we got wrong, and how we know. The sentence "one qutrit, self-entangled as past/future, is enough to generate the full substrate" is *not* ours. It is stated verbatim in this project's own encyclopedia, `docs/index.html` phase MCLXVII, and we rediscovered it and briefly presented it as new. What survives as new is narrower: the explicit construction of the 40 points as projective left×right multiplication classes on $\mathrm{End}(\mathbb{C}^3)$, and the identification of the 40 lines as the maximal *commuting* subalgebras. The lesson — search for the *result*, not the topic — produced three of the tools in §17.

4 The substrate: $W(3,3)$

In plain language. *Here is what the 40 directions look like when you write down which of them are compatible with which.*
Two measurements are compatible *if you can perform both without one disturbing the other. Of the 40 directions, each is compatible with exactly 12 others. Whenever two directions are compatible, exactly 2 further directions are compatible with both. Whenever two are* in*compatible, exactly 4 are compatible with both.*
Those four numbers — 40, 12, 2, 4 — are the shape's fingerprint. They are not *its name.*

An earlier draft of this page said the opposite. This section used to read: *"those four numbers pin the shape down completely; there is essentially one object in the world with that compatibility pattern."*

That is false, and this project's own encyclopedia says so in plain words: **Spence classified exactly 28 non-isomorphic graphs** with parameters $(40, 12, 2, 4)$. Two of them are the ones we care about. Quoting the parameters identifies nothing; the object is fixed by its *construction*.

The error mattered, because §10 proves a headline result by computing these parameters and concluding the object was $W(3, 3)$. That inference was one step short. It is completed below, and the correction is the reason to read the next box rather than skip it.

How the object is actually pinned down, Pass 2869. Three facts in sequence, each cheap and each exact:

1. **Parameters** $(40, 12, 2, 4)$ — narrows the field to 28 graphs.
2. **The quadrangle axiom**: every edge lies in exactly one 4-clique (there are 40 of them, four through each point), and for every such line ℓ and every point off it, *exactly one* point of ℓ is adjacent. Verified over all 40 lines. This narrows 28 to **two**: $W(3, 3)$ and its dual $Q(4, 3)$, which are not isomorphic at odd q.
3. **A spread exists** — ten pairwise disjoint lines covering all forty points, exhibited explicitly. $W(3, q)$ always has one; $Q(4, q)$ has one only for *even* q. At $q = 3$ this settles it.

In plain language.*So the shape does have a name, and it has been studied since the 1900s: $W(3, 3)$. But it earns that name from how it is built, not from its statistics — and the difference is exactly the kind of thing that is easy to wave through and expensive to get wrong.*

4.1 The definition, stated once and precisely

In plain language.*Everything downstream depends on this object being the right one, so here it is written out — and it is worth noticing how short the definition is. The whole substrate is one line of arithmetic.*

$W(3, 3)$, in full. $W(3, 3)$ is the **symplectic polar space** $W(3, \mathbb{F}_3)$.

Points. The 40 projective 1-spaces of $\mathrm{GF}(3)^4$. Every 1-space is automatically isotropic for an alternating form, so *all* of them are points — there is nothing to exclude.

Adjacency. With J the standard symplectic form on $\mathrm{GF}(3)^4$,

$$\boxed{v \sim w \iff v^\top J\, w \equiv 0 \pmod 3}$$

That single congruence generates every number in this document.

Lines. The 240 edges pair up into 40 *totally isotropic projective lines*: each edge is a pair of orthogonal points spanning exactly one such line. A *non*-edge spans an ordinary hyperbolic line instead — the two kinds of pair are geometrically different objects, not merely present-or-absent.

Graph. The collinearity graph is $\mathrm{SRG}(40, 12, 2, 4)$ with eigenvalues 12, 2, -4; it has **diameter** 2 and is **Ramanujan**.

Symmetry. $\mathrm{Aut} = \mathrm{Aut}(\mathrm{PSp}(4, 3)) \cong \mathrm{PGSp}(4, 3) \cong W(E_6)$, order 51,840, with normal inner subgroup $\mathrm{PSp}(4, 3)$ of order 25,920.

The same order, twice, for two different groups. $\mathrm{Sp}(4,3) = 2.U_4(2)$ also has order 51,840 and is **not** the group above. It has a centre C_2; the automorphism group has trivial centre. One is a central double cover, the other an outer extension, and they are not isomorphic.

This is the most common error in this area and this project has made it. Both groups appear here for good reasons — $\mathrm{Sp}(4,3)$ acts on Pauli frames and is what the hardware implements (§6); $\mathrm{PGSp}(4,3)$ acts on the drawing. Whenever 51,840 appears in this document, which group is meant is stated.

In plain language. *Two properties in that box are worth translating, because they are the ones an engineer should care about.*
Diameter 2 *means any point reaches any other in at most two steps. For an address space that is the strongest possible statement short of everything being adjacent to everything: no packet is ever more than one relay from its destination* within a node.

4.2 How good is "optimal", exactly?

In plain language. *"Ramanujan" is usually a tight call — a graph scrapes under the bound and earns the name. This one does not scrape.*

***Prior art:* `docs/index.html`.** *The non-trivial eigenvalues are $r = 2$ and $s = -4$, against a Ramanujan bound of*

$$2\sqrt{k-1} = 2\sqrt{11} \approx 6.63,$$

so both clear it by a factor of more than 1.6. *The spectral gap is* 10.

And it satisfies a Riemann Hypothesis.

The Ihara zeta function. *Every non-trivial pole of $\zeta_{W(3,3)}(u)$ lies* exactly *on the critical circle $|u| = 1/\sqrt{11}$:* 48 *poles from $r = 2$,* 30 *from $s = -4$, for*

$$48 + 30 = \mathbf{78} = \dim(E_6)$$

complex poles in total. That is the graph-theoretic Riemann Hypothesis, and this graph satisfies it. The pole discriminants encode the graph back: $|\operatorname{disc} r| = 40 = v$, $|\operatorname{disc} s| = 28 = v - k$, and their difference is $k = 12$.

In plain language. *For an engineer the practical content is the first box: information crosses this network as fast as information can cross any network of this size and degree, with a wide margin rather than a technicality. The second box is the reason — the same reason a Riemann Hypothesis is about anything at all, which is that the object's spectrum is as tightly controlled as a spectrum can be.*
It also sharpens §16's comparison considerably. The address layer does not merely beat the instruction layer; it is provably at an optimum the instruction layer misses.

Ramanujan *is a technical optimality condition on the eigenvalues. Informally, it means the graph is as well-connected as a graph of that degree can possibly be — you cannot build a better-mixing network on* 40 *nodes with* 12 *links each. The routing fabric is not merely adequate; it is extremal.*

4.3 What the shape already contains

In plain language. *The definition is one congruence. What follows from it is not obvious, and most of the machine is downstream of these four facts rather than designed.*

Homology: an exact 81-sector. The simplicial chain complex of the collinearity graph gives

$$H_1\big(W(3,3);\mathbb{Z}\big) = \mathbb{Z}^{81}.$$

That is the same dimension as $\mathfrak{g}_1$ in the $\mathbb{Z}_3$-graded decomposition of the exceptional Lie algebra E_8,

$$E_8 = \mathfrak{g}_0(86) \oplus \mathfrak{g}_1(81) \oplus \mathfrak{g}_2(81), \qquad \mathfrak{g}_0 = E_6 \oplus A_2.$$

Hodge theory: four sectors and an exact gap. The Hodge Laplacian L_1 on 1-chains has exactly four eigenvalues:

eigenvalue	multiplicity	sector
0	81	harmonic — represents H_1
4	120	
10	24	
16	15	

The spectral gap $\Delta = 4$ is exact and separates the harmonic sector from the rest.

It is not a mass, and not a Yang–Mills gap. Either reading would need a curved four-dimensional bridge and an explicit field identification, neither of which this document has. It is a finite eigenvalue of a finite matrix.

The substrate is its own error-correcting code. The clique complex of $W(3,3)$ over GF(3) is an exact qutrit CSS code

$$[240,\ 81,\ d_Z = 4],$$

with check ranks 39 and 120 and an explicit weight-4 logical cycle. So the 240 wires *are* the code, the 81 logical values *are* the frame space's dimension count, and no separate error-correction layer has to be bolted on.

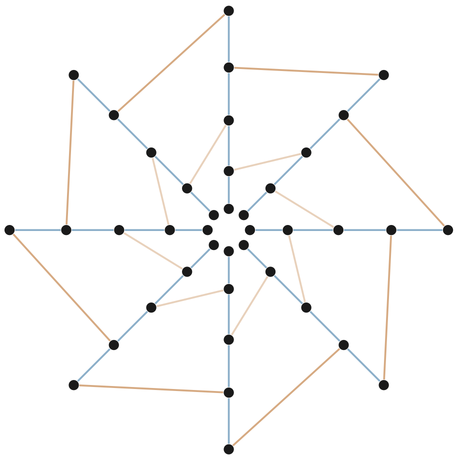
Euler characteristics, because two circulate and they are both right about different things: the truncated shell gives $40 - 240 + 160 = -40$, and the full tetrahedral complex $(40, 240, 160, 40)$ gives -80.

The 240-to-E_8 temptation. The 240 edges factor as $40 \times 3 \times 2$ (lines $\times$ matchings $\times$ edges per matching), and E_8 has exactly 240 roots, which under $E_8 \to E_6 \times SU(3)$ branch as $3 \times (24 + 2 + 27 + 27)$. An E_8 Dynkin subgraph even sits inside the adjacency graph directly, at vertices $[7, 1, 0, 13, 24, 28, 37, 16]$, whose Gram matrix $2I - A_{\text{sub}}$ has determinant 1 — the E_8 Cartan matrix exactly.

And there is no equivariant bijection between the 240 edges and the 240 roots. The inner group PSp$(4, 3)$ is edge-transitive, so the edges form a single orbit; the correspondence is representation-theoretic, not lattice-geometric. This project spent several passes on the stronger reading before establishing that it is false, and the finding is recorded in §19.

In plain language. *That last box is the pattern this whole document is organised around. The shape produces number after number that matches something famous. Some of those matches are maps and some are arithmetic, and the only way to tell is to check each one — which is why the ledger in §?? has a status column and why §19 is as long as it is.*

4.4 Why this shape and not another



40 **points,** 40 **lines.**

Every point lies on exactly 4 lines. Every line carries exactly 4 points. Given a line ℓ and a point $P \notin \ell$, there is *exactly one* point of ℓ collinear with P.

(Schematic. The full incidence graph has 160 flags and does not embed in the plane without crossings; this figure shows the ring structure and two of the line families only.)

In plain language. *The "exactly one" statement is an incidence projection rule: for a point outside a line there is one point on that line collinear with it. It does not imply unique shortest paths in the collinearity graph. Each four-point line is a K_4, so that graph contains triangles; the triangle-free object is the bipartite point–line incidence graph, whose generalized-quadrangle girth is eight. Routing claims must therefore be proved by the routing code, not inferred from the word quadrangle.*

4.5 The wiring as a budget

In plain language. *One more fact about the shape, and it is the kind that makes a reader sit up — so it comes with an unusually careful statement of what it does* not *mean.*
Treat the 240 *connections as a physical system and ask how they vibrate. The answer splits them into four groups.*

Hodge Laplacian on 1-chains, recomputed independently, Pass 2884.

$$240 = \underbrace{81}_{\text{harmonic}} + \underbrace{120}_{\lambda=4} + \underbrace{24}_{\lambda=10} + \underbrace{15}_{\lambda=16}$$

The harmonic sector has dimension 81 — the frame register file, and the logical dimension of the substrate's own qutrit code $[240, 81, d_Z{=}4]$.

What is tempting here, and why it is not claimed. Two of the other three numbers are already architectural quantities in this document: 15 is the dimension of the support shell, and 24 is the order of the single-qutrit Clifford group modulo phase. It is very tempting to read the line above as "the machine's wiring decomposes into its own subsystems."

No map is exhibited in either case. These are equal integers. This project's second most expensive failure mode is precisely an over-read of a count match — and §6 contains a worked example where exactly this kind of coincidence, tested, turned out to be nothing. The decomposition is recorded as an open question: is either coincidence carried by a natural map? Until someone answers that, it is arithmetic.

4.6 Why three?

In plain language. *A fair question, and the one a sceptical reader should ask: the machine uses three-way switches rather than two-way ones, and three-way switches are unusual. Was three chosen because it works, or does something choose it?*
The honest answer is that this project found five independent conditions that each single out three, none of them adjustable. Here are two anyone can check by hand.

Two of the five selection conditions. Counting. A geometry of this type with q-way switches has $q(q+1)^2(q^2+1)/2$ edges. The number of elements you gain by extending the arithmetic five steps is $q^5 - q$. Demanding these agree:

$$q^5 - q = \frac{q(q+1)^2(q^2+1)}{2} \implies 2(q-1) = q+1 \implies \boxed{q=3} \text{ uniquely.}$$

Checked directly: $q^5 - q$ for $q = 2, 3, 4, 5, 7, 11$ gives 30, **240**, 1020, 3120, 16800, 161040, while the edge counts are 45, **240**, 850, 2340, 11200, 96624. They meet only at three.

Symmetry. The symmetry group of the geometry and the symmetry group of the exceptional structure E_6 have the same order — 51,840 — only at $q = 3$.

$240 = 3^5 - 3$. **The same** 240 **is the edge count of the geometry, the number of roots of** E_8**, and the count of non-base elements in the degree-five extension of the three-element field. Three conditions, one number, no free parameters.**

All five, for completeness.

#	condition	forces
1	$q^5 - q$ equals the $GQ(q,q)$ edge count	$q = 3$ uniquely
2	$\sin^2\theta_W = 3/8$ at unification	$3q^2 - 10q + 3 = 0 \Rightarrow q = 3$
3	K_{q+1} has exactly q perfect matchings	only $q = 3$: K_4 has 3
4	non-neighbours = dim(E_6 fundamental)	$v - k - 1 = q^3 = 27 \Rightarrow q = 3$
5	$\lvert\mathrm{PGSp}(4,q)\rvert = \lvert W(E_6)\rvert$	the order equality selects $q = 3$

Every one has q as an integer unknown and exactly one positive solution. Prior art: `docs/index.html` owns all five.

In plain language. *Three more conditions in the same family — involving the electroweak mixing angle at unification, the number of perfect matchings of a small complete graph, and the dimension of the fundamental representation of E_6 — also give three and only three. None of them was fitted; q appears in each as an integer to be solved for, and each equation has exactly one positive integer solution.*

That does not make three necessary *in any deep sense we can prove. It does mean the machine's most unusual design choice is not a preference.*

4.7 The two groups of order 51,840, which are not the same group

Exactly. Two distinct groups of order 51,840 appear in this project, and confusing them is the most common error in this area:

$$\mathrm{Sp}(4,3) = 2.U_4(2) \quad \text{(centre of order 2)}, \qquad \mathrm{PGSp}(4,3) = U_4(2){:}2 = W(E_6) \quad \text{(trivial centre)}.$$

They are *not* isomorphic. The first is the group acting on Pauli frames — the one the hardware implements. The second is the automorphism group of the geometry — the one the drawing has. One is a central double cover; the other is an outer extension.

In plain language. *This distinction is not pedantry, and it has physical content. The extra element in the first group acts as a sign, $z \mapsto -1$. Whether that sign is "spent" or "free" decides whether the resulting structure has a handedness — whether it distinguishes left from right. This is why the machine can* host *a chirality, though as we established in earlier work it cannot* explain *one from the inside.*

5 Four representations, four jobs

In plain language. *The machine now has four compact descriptions of the same underlying frame, and they must not be confused. The seven-bit code is the execution state. The four-bit support mask is cheap one-way telemetry. The protected observer words trade more wires and time for error correction. The ninety-six-state selector–tomotope token is a control word, not stored quantum data. Earlier drafts occasionally used the word "state" for all four; this section removes that ambiguity from the architecture.*

Representation contract.

Affine frame 7 bits. Lossless for all 81 Pauli frames; execution may read and write it.

Support telemetry
4 bits. Zero/nonzero mask only, and frame → support is one-way in the netlist — proved at the gate level, not asserted.

Protected observer
24–28 taps. Diagnostic word for identification and correction; never the live datapath.

Control token
7 raw bits, 96 valid. Selects face, matching channel and phase cell.

Support is observable, not executable. The refinement count $16 \to 40 \to 78 \to 81$ was once easy to read as four increasingly accurate compressed machine states. It is not. Exhausting every nonempty subset of the four-operation micro-ISA proves that the coarsest exact support-refining execution quotient for the full ISA has all 81 singleton classes. The intermediate partitions are observer states only. In fact F_p, Z_p, and either controlled-add direction already force the full frame.

Intrinsic selector channel. For a noncollinear pair x, y and a distinguished common neighbour c, the three residual selector channels form the affine line $C(x, y)\setminus\{c\}$. The multiplier-one stabilizer induces its translation group C_3; the full projective similitude stabilizer induces $\mathrm{AGL}(1, 3) \cong S_3$. There is therefore no globally equivariant numbering $0, 1, 2$. Hardware may choose a local encoding, but interfaces transport it as an affine-line torsor rather than pretending the labels are geometric invariants.

Part II — The machine

What the geometry forces the instruction set, the datapath and the virtualisation layer to be — measured on real silicon throughout.

6 The instruction set

6.1 The idea of a Pauli frame

In plain language. *Here is the trick that makes the machine cheap.*
A quantum state is expensive to write down: for two qutrits you need nine complex numbers, and keeping them accurate is most of what makes quantum computing hard. But for a large and useful family of operations — the Clifford *operations — you never need the state at all. You only need a* label *saying how the state has been shifted relative to where it started.*
That label is four trits. Four trits is eight bits. The machine's entire "register file" for tracking Clifford operations is one byte.

The frame unit tracks a label, not a state. Four trits, 81 possible values, eight bits of memory — and it correctly accounts for a group of 4,199,040 distinct operations.

In plain language. *That* 81 *is worth pausing on, because it turns up somewhere it has no obvious business being.*
Treat the geometry as a shape and ask a purely topological question — how many independent loops does it have? The answer is that its first homology group is exactly $\mathbb{Z}^{81}$*: eighty-one independent cycles. Nothing about that computation mentions registers, instructions, or qutrits. It is the same number as the size of the machine's register file.*

The coincidence, and the answer.

$$H_1\big(W(3,3);\mathbb{Z}\big) = \mathbb{Z}^{81}, \qquad |\mathbb{F}_3^4| = 81.$$

Both are 3^4, and the shared exponent is not an accident: the geometry lives in a four-dimensional space over a three-element field, and both counts are governed by that. The clique complex over $\mathbb{F}_3$ is moreover an exact qutrit error-correcting code with parameters $[240, 81, d_Z{=}4]$ — so the same 81 is also the number of *protected* logical values the substrate carries.

But the two spaces are not the same object (Pass 2883). Ask whether the symmetry group sees them alike: an order-three linear opcode fixes exactly 27 of the 81 frames, while the homology character on every order-three class is 0. The characters disagree, so there is *no* equivariant isomorphism. The counts match for a real reason and the spaces still differ — which is the most common shape a numerical coincidence takes in this project, and the reason every one of them gets this test.

In plain language. *It would have been more satisfying if the answer were yes. Writing down the negative result is the point of having the test at all: an earlier draft of this page left the question open, which reads as promising, and open questions that stay open too long start to be cited as though they were answered.*

Exactly. The Pauli frame is $(x_p, z_p, x_f, z_f) \in \mathbb{F}_3^4$, denoting the operator $X^{x_p}Z^{z_p} \otimes X^{x_f}Z^{z_f}$. A Clifford unitary U acts on frames by conjugation, $U(X^aZ^b)U^\dagger = X^{a'}Z^{b'}$ up to phase, and the induced map on $\mathbb{F}_3^4$ is *linear and symplectic*: it preserves

$$\langle u, v\rangle = (x_pz_p' - z_px_p') + (x_fz_f' - z_fx_f') \bmod 3.$$

6.2 The eight opcodes

$\mathcal{I}_{\text{holo}}$, **complete semantics.**

Op	Name	Action on (x_p, z_p, x_f, z_f)	Kind
000	F_p	$(x_p, z_p) \mapsto (-z_p,\ x_p)$	linear
001	F_f	$(x_f, z_f) \mapsto (-z_f,\ x_f)$	linear
010	S_p	$z_p \mapsto z_p + x_p$	linear
011	S_f	$z_f \mapsto z_f + x_f$	linear
100	CX	$d{=}0$: $z_p \mapsto z_p - z_f,\ x_f \mapsto x_f + x_p$	linear
		$d{=}1$: $x_p \mapsto x_p + x_f,\ z_f \mapsto z_f - z_p$	linear
101	$\sigma^5{=}Z$	$z_p \mapsto z_p + 1$ or $z_f \mapsto z_f + 1$	**affine**
110	D_{12}-mirror	transport in the dihedral group D_{12}	(not a frame op)
111	M_{36}-magic	typed request for one of 36 Witting rays	(handshake)

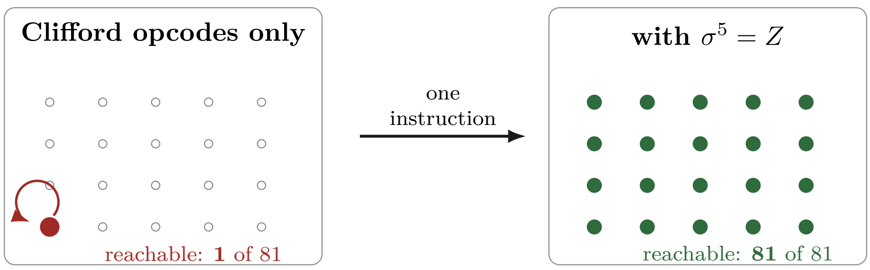
6.3 The reachability theorem, and why it matters

In plain language. *Now a question that sounds trivial and is not:* starting from a blank machine, which states can these instructions actually reach?

The answer is startling. All six of the Clifford instructions, applied in any order, any number of times, reach exactly one state *— the blank one. They cannot move the machine at all.*

The reason is simple once you see it. Those six instructions are all linear, *and a linear map always leaves the origin where it is. Multiplying zero by anything gives zero. So a machine built from only those six instructions is, quite literally, a machine that does nothing, no matter how sophisticated its circuitry looks.*

The seventh instruction, $\sigma^5 = Z$, is different: it adds one. *Addition moves the origin. That single instruction is what makes all* 81 *states reachable — and hence what makes the machine a computer instead of an ornament.*



Proved by exhaustive search, Pass 2774. Breadth-first search over all $3^4 = 81$ frames from $(0,0,0,0)$:

CX alone	1
all six symplectic opcodes	1
full ISA (adding $\sigma^5{=}Z$)	81

Reproduce: `py -3 analysis/w33_pass2774_isa_frame_reachability.py`
Certificate: `data/PART_W33_PASS2774_ISA_FRAME_REACHABILITY.json`

6.4 How much computation the eight opcodes actually buy

In plain language. *"All 81 states are reachable" is weaker than it sounds — it says you can get anywhere, not that you can perform every operation. The stronger question is: what is the complete list of things this instruction set can do?*

Proved, Pass 2778. The six Clifford opcodes generate a group of order exactly 51,840 — that is, the whole of $\mathrm{Sp}(4,3)$, nothing missing. Adjoining the translation gives

$$\langle \mathcal{I}_{\mathrm{holo}} \rangle \;=\; \mathrm{ASp}(4,3) \;=\; \mathbb{F}_3^4 \rtimes \mathrm{Sp}(4,3), \qquad |\cdot| = 81 \times 51840 = 4{,}199{,}040.$$

One translation generator suffices. Enabling only Z on the past register gives the same group as enabling both, because $\mathrm{Sp}(4,3)$ is transitive on the 80 nonzero frame vectors: the orbit of a single translation already spans $\mathbb{F}_3^4$. Measured orbit size: 80. A second translation instruction adds nothing and can be removed from the ISA.

Eight opcodes. Three bits. 4,199,040 **distinct operations, exactly — and the eighth opcode is provably redundant.**

6.5 How small can the instruction set be?

In plain language. *If one opcode is redundant, how many are? This is not idle: every instruction removed is decoder logic removed, and if enough go, the opcode field itself gets shorter — which shortens every instruction word in the machine.*
The answer is that you can throw away half of them.

Exhaustive over all subsets, Pass 2789. Because $\mathbb{F}_3^4 \rtimes H = \mathrm{ASp}(4,3)$ iff $H = \mathrm{Sp}(4,3)$ and at least one translation is present, the search reduces to the six linear opcodes:

subsets of size 1 generating $\mathrm{Sp}(4,3)$:	0
subsets of size 2 :	0
subsets of size 3 :	**6**

The six minimal triples are $\{F_p, F_f, \mathrm{CX}_{p\to f}\}$, $\{F_p, F_f, \mathrm{CX}_{f\to p}\}$, $\{F_p, S_f, \mathrm{CX}_{p\to f}\}$, $\{F_p, \mathrm{CX}_{p\to f}, \mathrm{CX}_{f\to p}\}$, $\{F_f, S_p, \mathrm{CX}_{f\to p}\}$, $\{F_f, \mathrm{CX}_{p\to f}, \mathrm{CX}_{f\to p}\}$.

Three Clifford opcodes plus one translation generate everything. Four frame instructions — a two-bit opcode field, not three.

In plain language. *Six choices, all valid — so pick whichever is cheapest to build, and the decision costs nothing. That was the obvious inference, and it is wrong.*

The six triples do *not* cost the same, Pass 2882. Exhaustive breadth-first search over the full group, once per triple:

triple	diameter	mean
$F_p + \mathrm{CX}_{p\to f} + \mathrm{CX}_{f\to p}$	**19**	**14.18**
$F_f + \mathrm{CX}_{p\to f} + \mathrm{CX}_{f\to p}$	20	15.22
$F_p + F_f + \mathrm{CX}_{f\to p}$	22	16.08
$F_p + S_f + \mathrm{CX}_{p\to f}$	22	16.12
$F_f + S_p + \mathrm{CX}_{f\to p}$	22	15.90
$F_p + F_f + \mathrm{CX}_{p\to f}$	23	17.08

Worst-case program length varies from 19 to 23 across the six — a 21% spread for a choice that looks free. Build the one with two entanglers.

In plain language. *The pattern is legible once seen: the best triple is the one containing* both *directions of the coupling instruction, and the worst is the one containing neither. What shortens programs is the ability to move information both ways between past and future, which is a satisfying thing for the machine's own central metaphor to be rewarded by its compiler.*

6.6 How long is the longest program?

In plain language. *"These four operations can do everything" is a statement about what is* possible*. The question a compiler actually asks is what it* costs*: if I hand you an arbitrary frame transformation, how many instructions do you need in the worst case?*
That number is the machine's diameter*, and it can be computed exactly, because the group is only four million elements and a computer can visit all of them.*

Exhaustive breadth-first search over $\mathrm{ASp}(4,3)$**, Pass 2866.** Forward generators only — a processor cannot run an instruction backwards, so the *directed* diameter is the honest number.

$$\boxed{\text{diameter} = 19}, \qquad \text{mean program length} = 14.176, \qquad \text{counting floor } \lceil \log_4 4199040 \rceil = 12.$$

The last shell is tiny: only 188 of 4,199,040 elements are at distance exactly 19. Ball sizes reach 93.3% at depth 16 and 99.996% at depth 18.

Every frame transformation is at most 19 instructions away, and the average is 14.2. No four-operation instruction set can beat 12, so this one is worst-case optimal to within a factor of 1.73.

In plain language. *The method is worth a sentence because it is what makes the computation possible at all. A group element is a matrix together with a shift. Each instruction acts on the matrix part as a fixed* shuffle *of* 51,840 *labels and on the shift part as a fixed map of* 81 *labels. Build those two lookup tables once, and searching four million elements is just table lookups — seconds, not days.*

6.7 And how long until the machine forgets?

In plain language. *The opposite question. Feed the machine random instructions: how many before its state tells you nothing about where it started?*

Exact, Pass 2867. Every opcode is a bijection on the 81 frames, so the walk is doubly stochastic and the stationary distribution is exactly uniform — *no frame is preferred.*

$$|\lambda_2| = 0.893992, \qquad \mathrm{gap} = 0.106008, \qquad \frac{1}{\mathrm{gap}} = 9.43, \qquad t_{\mathrm{mix}}(\tfrac{1}{4}) = 15, \qquad K = 113.38.$$

A frame forgets its origin by $1/e$ in 9.4 instructions and is statistically gone after 15 — 71.8 ns at the measured 208.86 MHz.

And what the worst case actually looks like, Pass 2885. Only 188 of 4,199,040 elements sit at distance exactly 19 — 0.0045%. The natural guess is that so rare a set is one orbit, one coset, something nameable.

It is not. Those 188 spread over 54 distinct linear parts, element orders 4 through 18, and all three trace classes. There is no single "hardest instruction" to name, and the structural hypothesis is refuted.

What survives is the useful half: 188 **is small enough to enumerate exactly**, so a compiler regression suite can contain *every* worst-case input rather than sampling for them. A negative structural result and a positive engineering one out of the same run.

In plain language. *Put that beside the readout cost of §11 and you get an observation cadence, which is a real design parameter: reading more often than every fifteen instructions pays the thermodynamic price repeatedly for information the machine has not yet regenerated.*

Notice also how close 15 *and* 19 *are. The machine scrambles almost exactly as fast as it can reach — there is no regime in which it is still reachable but already random, which is a comfortable property for a scheduler to have.*

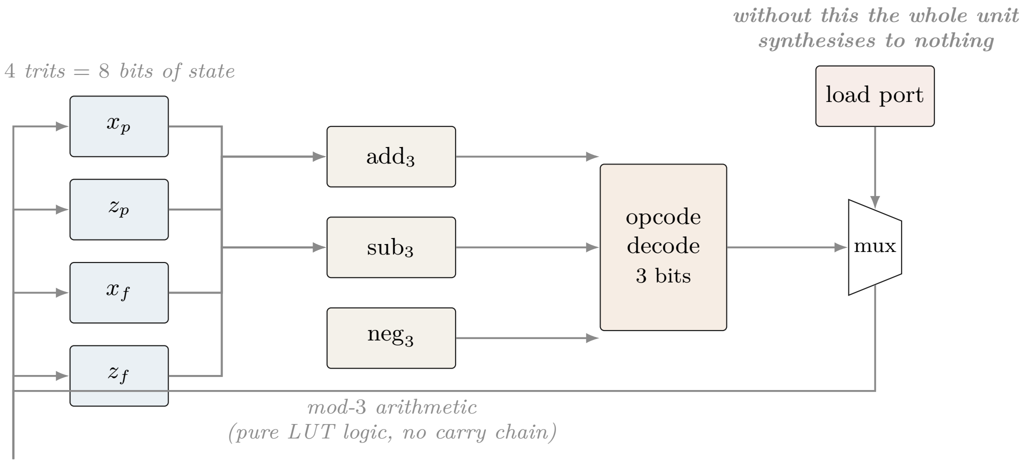
A trap avoided, and worth naming. A minimality *proof* says nothing about *area.* It is tempting to quote the 72-cell figure for the four-operation engine on the grounds that it does strictly less work — and that would be wrong twice over: the number was measured on a different design, and removing decode cases can *cost* cells when what was removed was being shared with what remains. The minimal engine was therefore synthesised and placed separately (§16), and only then quoted. It came out at 43 cells; had it come out at 80, that is the number this document would print.

In plain language. *Two details are worth noticing. First, there are* six *different minimal choices, so a chip designer has real freedom to pick whichever three are cheapest in silicon rather than being handed one answer.*

Second, every single one of the six contains at least one Fourier gate and *at least one controlled-add. No collection of phase gates and one Fourier gate will do. The entangler is not optional — which is the formal version of the intuition that a machine that never couples past to future is not really using the self-entanglement at all.*

7 The hardware

7.1 The frame unit datapath



`rtl/w33_pass2777_isa_frame_unit.sv`

7.2 The measured cell budget

In plain language. *Below is what the machine actually costs, measured by running the industry-standard open-source tools — Yosys to translate the design into logic gates, nextpnr to place those gates on a real chip — targeting a Lattice iCE40 UP5K, a chip you can buy on a breakout board for about $25.*

The entire arithmetic core of this computer is seventy-two *logic cells, out of* 5,280 *available. It uses* 1.4% *of a $25 chip.*

The three designs, on two parts, in one harness.

	HX8K CT256		UP5K SG48		
Design	**LC**	$F_{\max}$	**LC**	$F_{\max}$	**Opcode**
Minimal engine, 4 operations	**43**	**208.86**	43	72.40	2 bits
Public unit, 6 frame opcodes	72	175.72	72	60.80	3 bits
Engine + support readout	48	207.64	48	72.05	2 bits

frequencies in MHz

Same cells on both parts — synthesis does not know the package — but the HX8K is 2.9× faster. **Every UP5K figure elsewhere in this document therefore understates speed by about a factor of three**, and is kept because it is the part the $25 board carries.

Measured on UP5K SG48; the per-instruction breakdown.

Design	Cells	Pins	$F_{\max}$	Opcode
Minimal engine, 4 operations	**43**	22	**72.40** MHz	2 bits
Public unit, 6 frame opcodes	72	26	60.80 MHz	3 bits

40% **fewer cells and** 19% **faster.** The minimality theorem cashes out in silicon. Also verified on the minimal engine, because both checks are cheap and both have caught real defects here: the linear part is symplectic on frame *differences* (SAT, 4100 variables / 11278 clauses — a translation cancels in a difference, so one assertion covers all four opcodes at once), and no output folds to a constant.

Measured; iCE40 UP5K SG48, Yosys 0.67 + nextpnr.

Build	Logic cells	$F_{\max}$ (MHz)	Contains
F only	16	230.84	one Fourier gate
$F_p + F_f$	19	230.84	both Fourier gates
F and S (4 opcodes)	40	86.60	the single-qutrit Cliffords
CX only	33	72.40	the entangler, both directions
σ^5=Z only	20	144.07	the translation
all Clifford, no Z	65	60.80	cannot leave the origin
full frame unit	**72**	**60.80**	the machine
For comparison, elsewhere in the design:			
CX, bare loadable	23	72.40	`w33_pass2752_cx_loadable_frame.sv`
36-lane mixer, serial	4048	19.65	`w33_pass2612_serial_mixer.sv`
36-lane mixer, parallel	13,965 LUT4 + 799 carry		does not fit any iCE40

In plain language. *The last two rows are worth a moment. The "mixer" is the part that combines* 36 *signals at once. Built the obvious way — all* 36 *lanes in parallel — it needs about* 14,000 *logic cells, nearly twice what the largest chip in this family has. Built serially, one lane at a time, it needs* 4,048*: about a third of the area, at* 1/36 *of the speed. That is the kind of trade a blueprint exists to make explicit.*

What we got wrong, and how we know.Every cell count in this project before Pass 2753 was wrong, and the reason is instructive. We reported 13 cells for F and S, 8 for CX, 42 for the ISA. Those modules had no load port. By the reachability theorem of §6, a register driven only by Clifford opcodes can never leave $(0,0,0,0)$ — so the synthesis tool *proved the state was constant and deleted six of the eight flip-flops.* We were measuring the area of nothing.

Two independent teams shipped this same defect within four hours of each other. Both had *exhaustive* testbenches — one covered all 81^2 pairs of frames — and both passed, because the testbenches drove the *combinational* logic and never instantiated the sequential wrapper.

Simulation cannot detect this class of defect at all. A module that folds to a constant still simulates correctly. It just does nothing. Only a synthesis frontend asks whether the state can move.

One figure survived unchanged: $\sigma^5 = Z$ was reported at 21 cells and re-measures at 20. It was always real — because it is the one instruction that is not linear, and so the one instruction that could always move the state. The theory predicted exactly which number would survive.

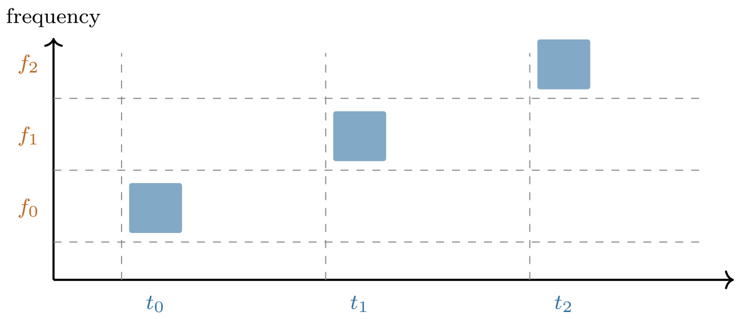
Part III — The physics

Light, magic states, thermodynamics: where the machine touches the world, and exactly how far the evidence goes.

8 The physical layer

In plain language. *Everything above is logic — it would work on any hardware that can implement the eight operations. This section is about doing it with light, which is the intended implementation, and about what has already been demonstrated in a laboratory as opposed to designed on paper.*

A single photon carries several independent properties. Two of them are useful here: when *it arrives (its time bin) and* what colour *it is (its frequency bin). Chop time into three slots and colour into three bands, and one photon carries two qutrits — the "past" and "future" registers of §3.*



The sum gate

$|f, t\rangle \mapsto |f,\ t + f \bmod 3\rangle$

A chirped fibre Bragg grating delays each colour by a different amount, so the photon's *colour* controls how far its *arrival time* shifts. That is a controlled-add, implemented with one passive optical element.

Demonstrated: Imany *et al.*, *npj Quantum Information* **5**, 59 (2019). Three 3 ns time bins at 6 ns spacing, three frequency bins at 380 GHz, −2 ns/nm dispersion, 18 ns wraparound. Measured basis fidelity 0.92 ± 0.01; certified entanglement of formation $\geq 1.19 \pm 0.12$ ebits.

Evidence boundary — what is measured and what is not. Measured in a laboratory: the deterministic single-photon qutrit SUM gate, at fidelity 0.92 ± 0.01, with the entangled output certified.

Proved exactly: the group theory, the geometry, the class atlas, the ISA semantics, the reachability, the synthesis and placement figures.

Neither, and stated as such: source efficiency, insertion-loss engineering, fault-tolerance thresholds, and the magic-state injection route. The M_{36} resources are typed `M36_Q4_RAW` in the controller and injection is *refused* in RTL until a proved ququart protocol supplies a validity assertion. That refusal is a design feature: it makes the gap impossible to paper over.

9 Why the exponent nine

In plain language. *A small but pretty result that shows how tightly the layers fit.*

To tell which operation the machine just performed, you measure a quantity and look it up in a table. The obvious quantity — the "trace" — has a flaw: it changes if the whole state picks up an irrelevant overall phase, which physically means nothing. The fix used in this project is to raise the trace to the ninth *power and divide by a determinant. The ninth power was found empirically. Where does nine come from?*

Derived, Pass 2779. For a d-dimensional representation, under $U \mapsto \lambda U$ we have $\mathrm{Tr}(U^k)^e \mapsto \lambda^{ke}\,\mathrm{Tr}(U^k)^e$ and $\det(U^k) \mapsto \lambda^{kd}\det(U^k)$, so

$$\Theta_k(g) = \frac{\mathrm{Tr}(U_g^k)^e}{\det(U_g^k)}$$

is phase-invariant for all λ *exactly* when $e = d$. Here $d = 9$, the dimension of the two-qutrit state space — which by §3 is $\dim \mathrm{End}(\mathbb{C}^3)$. Verified numerically: $e \in \{3, 8, 10\}$ fail, $e = 9$ holds.

If the phase ambiguity were a *finite* group μ_m one could use any $e \equiv 9 \pmod{m}$, possibly smaller. We computed the scalar subgroup of the generated matrix group by collision sampling and it is μ_{12}. Since $9 \not\equiv 0 \pmod{12}$, the determinant is genuinely required, and 9 is the *smallest* exponent that works: the sensor is minimal.

$\mu_{12} = \mu_4 \times \mu_3$. The μ_3 is ω, the qutrit alphabet. The μ_4 is the quadratic Gauss sum $\sum_j \omega^{j^2} = i\sqrt{3}$, which is *imaginary* precisely because $3 \equiv 3 \pmod{4}$ — the same congruence that makes time reversal a physical rather than a gauge symmetry.

In plain language. *That last box is the kind of thing that makes this project worth doing. A practical question — "why is the exponent nine in the measurement circuit?" — turns out to have the same answer as an abstract one — "why does this geometry distinguish past from future?". Both are the statement that -1 is not a perfect square when you count modulo 3.*

9.1 And for wider registers, it is usually three

Pass 2791. The phase group is not an artefact of the two-qutrit case. At $n = 1$ the Clifford group is small enough to enumerate exactly rather than sample:

$$|\langle F, S, X\rangle| = 2592 = \underbrace{12}_{\mu_{12}} \times \underbrace{216}_{|\mathrm{ASp}(2,3)|} \qquad \text{— consistent, so } \mu_{12} \text{ again.}$$

For n qutrits $d = 3^n$, so the minimal exponent is the least $e \equiv 3^n \pmod{12}$, and 3^n mod 12 cycles $3, 9, 3, 9, \ldots$:

n	1	2	3	4
$d = 3^n$	3	9	27	81
minimal e	**3**	9	**3**	9

Verified directly on 3×3 unitaries: $e = 3$ is phase-invariant, $e \in \{2, 4, 9\}$ is not.

Odd register widths need only a *cube* of the trace, not a ninth power — a strictly cheaper measurement, alternating with width.

10 The magic states, and the protocol that was found

In plain language. *Everything described so far is, in a precise technical sense,* not enough*. The eight opcodes generate a large and useful group, but a machine restricted to those operations can be simulated efficiently by an ordinary computer. To get a genuine quantum advantage you need one more ingredient, traditionally supplied as a special prepared state called a* magic state*.*
The substrate hands us 36 *candidates for free —* 36 *special directions in the four-dimensional space, called the Witting rays. The question is whether those* 36 *can be* purified*: whether noisy copies can be distilled into cleaner ones. Without that, the machine is a very elegant classical simulator.*
This section is the one that changed most between drafts, and the sequence is worth following, because it is a small case study in how a negative result should be read. First an exhaustive search said no. Then an argument from resource accounting said the "no" could not be fundamental. Then a wider search said yes.
But before any of that: the 36 *are not a lucky accident. Count them against the machine's own address space.*

10.1 Magic is everything except one line

In plain language. *The machine addresses* 40 *points. There are* 36 *magic rays. That leaves* 4 *— and a* line *of this geometry contains exactly* 4 *points. Either a coincidence, or the whole story.*
Add the four missing directions back — they turn out to be the four coordinate axes, which is to say the ordinary, unmagical computational basis — and ask what the 40 *look like together.*

Exact, Passes 2835 and 2869. Adjoin $e_1, \dots, e_4$ to the 36 rays and compute the orthogonality graph on all 40:

$$\text{degrees } \{12{:}40\}, \qquad \lambda = 2, \qquad \mu = 4 \implies \mathrm{SRG}(40, 12, 2, 4).$$

Those parameters admit 28 non-isomorphic graphs, so the identification is completed by the two further steps of §4: the quadrangle axiom holds over all 40 lines (narrowing to $W(3,3)$ or $Q(4,3)$), and an explicit spread exists (which only $W(3,3)$ has at odd q). **The 40 rays are the points of $W(3,3)$.**

The four axes are mutually orthogonal — a 4-clique, hence a line. Every magic ray is orthogonal to exactly one axis ($\{1{:}36\}$), and the resulting nearest-point partition is $9+9+9+9$: precisely the four families the ray construction produces.

M_{36} is the 40 points of $W(3,3)$ with *one line* deleted,
and the deleted line is the computational basis.

In plain language. *This changes what the magic states* are*. They are not a resource bolted onto the substrate and supplied from outside. They are the generic condition of the machine's own address space: pick a point at random and it is magic with probability* 36/40*. The ordinary, classically-simulable states are the single distinguished line you have to delete to get rid of the magic.*
One more thing falls out, and it is a warning about reading structure off counts. The geometry partitions the 36 *into four families of nine. The Clifford group partitions them into* $4 + 8 + 12 + 12$*.* Neither refines the other. *The symmetry and the geometry are looking at different things.*

Is the deleted line special? Both answers are true, Pass 2862. Enumerating the 40 lines and counting how many classical points each contains:

$$\underbrace{1}_{\text{all four classical}} + \underbrace{12}_{\text{one classical, three magic}} + \underbrace{27}_{\text{all four magic}} = 40.$$

Geometrically, no. $\mathrm{Aut}(W(3,3))$ has order $51840 = 40 \times 1296$ and is transitive on lines, so nothing intrinsic to the 40 points prefers one. Choosing a computational basis is choosing one line out of forty — exactly $\log_2 40 = 5.3219$ bits of pure convention.

Relative to the stabilizer structure, uniquely yes. Exactly *one* of the forty lines has all four of its points classical.

In plain language. *Which is a satisfying place to end up. "Which states are magic" is not a fact about the geometry — the geometry is completely indifferent. It is a fact about which line you declared to be the basis. But the declaration is not lost afterwards: it is recoverable, because the line you chose is the unique all-classical one in the structure your choice created.*

10.2 All thirty-six look identical from the inside

Exact integer arithmetic in $\mathbb{Z}[\omega]$, Pass 2790. Only two overlap values occur among the 36 rays:

$$|\langle i|j\rangle|^2 \in \{\, 0,\ \tfrac{1}{3}\,\},$$

and every ray has the *same* profile: 11 orthogonal partners and 24 partners at overlap 1/3. There is exactly one Gram profile among all 36.

(The orthogonality graph is 11-regular on 36 vertices but *not* strongly regular: $\lambda \in \{1,2\}$, $\mu \in \{3,4\}$. Recorded because the near-miss invites a false claim.)

In plain language. *So no symmetry of the machine can tell one ray from another by looking at how they sit relative to each other. If the rays differ at all, they differ only in how they sit relative to the* ordinary *states — an external comparison, not an internal one.*

10.3 And from the outside, they fall into exactly three grades

60 **two-qubit stabilizer states,** $F_{\mathrm{stab}} = \max_s |\langle s|\psi\rangle|^2$**.**

F_{stab}	closed form	rays	grade
0.750000	$9/12$	**4**	shallow
0.705342	$(5+2\sqrt{3})/12$	**24**	mid
0.622008	$(4+2\sqrt{3})/12$	**8**	deep

Stabilizer fidelity is a Clifford invariant, so the $8 + 24 + 4$ grading is genuine and not an artefact of the chosen basis.

All three noise thresholds are one formula. For $\rho_p = (1-p)|m\rangle\langle m| + p\,I/4$ the target overlap is $1 - 3p/4$, so the witness certifies non-stabilizerness exactly while $1 - 3p/4 > F_{\text{stab}}$:

$$\boxed{p \;<\; \tfrac{4}{3}\big(1 - F_{\text{stab}}\big)}$$

$$\begin{array}{lll} \text{shallow} & 0.333333333333 & = \; 1/3 \\ \text{mid} & 0.392877598318 & = \; (7-2\sqrt{3})/9 \\ \text{deep} & 0.503988709429 & = \; (8-2\sqrt{3})/9 \end{array}$$

Three separately-published thresholds, one formula, agreement to 10^{-12}.

10.4 Four, not three: fidelity is necessary but not sufficient

What we got wrong, and how we know. An earlier draft of this section said: *"the distillation search needs three representatives, not thirty-six."* That was one step too far. Equal stabilizer fidelity does not imply the two states are interchangeable — it is a single number, and a single number rarely separates everything.

Acting with the full two-qubit Clifford group (92,160 matrices) on the 36 rays and matching images by *fidelity* rather than by a rounded key:

$$\text{Clifford equivalence classes} \;=\; 4, \qquad \text{sizes } [\,4,\ 8,\ 12,\ 12\,].$$

Every class sits inside one grade — so the grading is real — but the 24-ray middle grade is **two** inequivalent families of twelve. The correct count is **four**.

Corroborated from the other direction: an independent exhaustive distillation search, which knows nothing about Clifford orbits, reports results for "*both* middle grades". Two unrelated computations, the same splitting.

And nothing an overlap can see tells the twins apart, Pass 2830. The *full* stabilizer overlap spectrum — the multiset of $|\langle s|\psi\rangle|^2$ over all 60 stabilizer states — is a Clifford invariant, and every stabilizer Rényi entropy and overlap moment is a function of it. Computed exactly in ninths, the two 12-ray classes have the **identical** spectrum.

Across all four classes there are only **three** distinct spectra. So the entire 60-value distribution has exactly the same resolving power as its single largest entry, F_{stab}.

Consequence. A protocol that reads its input only through overlap statistics *must* treat the two classes alike. The split can matter only to a protocol that uses the ray's Clifford orbit — one that fixes a frame and cares which representative it was handed.

10.5 Why the first "no" could not have been the last word

In plain language. *An exhaustive search over* 5,355 *small codes and* 21,420 *branches found no two-copy protocol that improves fidelity. That is a real result, and the tempting reading is that two noisy copies simply do not contain enough raw material to make one better copy.*

There is a way to check that reading directly, without searching any protocols at all. Physicists have quantities that measure *how much magic a state contains and which are guaranteed never to increase under any allowed operation — no measurement, no post-selection, no clever feedback can make one go up. If two copies scored lower than the target, distillation would be impossible for reasons no ingenuity could overcome. So: compute the score.*

$D_{\min} = -\log_2 F_{\text{stab}}$ **over all** 36,720 **four-qubit stabilizer states.**

grade	$F_{\text{stab}}(\psi)$	$F_{\text{stab}}(\psi^{\otimes 2})$	$D_{\min}(\psi)$	$D_{\min}(\psi^{\otimes 2})$
shallow	0.750000000	0.562500000	0.415037	0.830075
mid	0.705341801	0.497507057	0.503606	1.007211
deep	0.622008468	0.386894534	0.684994	1.369988

F_{stab} is *exactly* multiplicative here, so $D_{\min}(\psi^{\otimes 2}) = 2D_{\min}(\psi) > D_{\min}(\psi)$ for every grade. The budget is never the binding constraint.

The monotone never obstructs. Any two-copy no-go in this system is *structural*, not information-theoretic — a fact about the protocol family searched, not about the resource. The right response to it is to widen the family.

10.6 And widening the family worked

Deep-grade distillation, exact. Extending the search from one decoder to all 11,520 logical Clifford decoders gives 48 **improving branches** on the deep eight-ray grade, and zero on the shallow and both middle grades. One explicit branch — input ray 5, stabilizers $IYZY$ and $YZXY$, syndrome $(-1, +1)$, output ray 7 — has the exact curve

$$P_{\text{succ}} = \frac{p^2 - 2p + 2}{4}, \qquad F_{\text{out}} = \frac{5p^2 - 12p + 8}{4(p^2 - 2p + 2)},$$

$$F_{\text{out}} - F_{\text{in}} = \frac{p(p-1)(3p-2)}{4(p^2 - 2p + 2)} \;>\; 0 \quad \text{for } 0 < p < \tfrac{2}{3},$$

an interval that contains the whole deep-grade magic window $p < (8 - 2\sqrt{3})/9 \approx 0.504$.

In plain language. *So the resource is usable. Note the shape of the story: the first search was correct and its conclusion was correctly stated; the accounting argument said the obstruction could not be fundamental; the wider search then found the protocol. Nobody was wrong at any step — but a reader who had stopped at "exhaustive search found nothing" would have drawn exactly the wrong conclusion about the machine.*

And now the part that a reader should not *stop before.*

10.7 Usable is not the same as practical

The convergence rate, Pass 2831. Linearising the accepted-round map at the pure fixed point:

$$\left.\frac{dp'}{dp}\right|_{p=0} = \frac{2}{3}, \qquad P_{\text{succ}}\Big|_{p=0} = \frac{1}{2}.$$

The noise is multiplied by two thirds each accepted round. It is **not** squared. Each round consumes two inputs and succeeds half the time, so the raw cost multiplies by $2/P_{\text{succ}} \approx 4$ per round while the infidelity only falls by a third.

start p	to 10^{-3}	to 10^{-6}	to 10^{-9}
0.10	2.3×10^{7}	4.0×10^{17}	6.9×10^{27}
0.20	5.6×10^{8}	9.7×10^{18}	1.7×10^{29}
0.30	1.5×10^{10}	2.6×10^{20}	4.5×10^{30}
0.50	1.1×10^{14}	1.9×10^{24}	3.2×10^{34}

raw noisy states consumed per distilled output

The map converges *linearly*. Standard distillation is quadratic or cubic — Reed–Muller 15-to-1 gives $p' \approx 35p^3$ — and that gap is the whole game.

In plain language. *So: the deep-grade branch is a genuine fidelity-improving map and a genuine refutation of the earlier no-go. It is* not, *by itself, a usable distillation routine. Ten to the twenty-seven raw states is not an engineering number.*
What it establishes is that the resource is not inert — which is the thing that was genuinely in doubt.

10.8 And no other two-copy branch will fix it

In plain language. *The natural next move is to hope one of the other 47 branches converges faster. It does not, and the reason can be settled without examining any of them.*
For a protocol to be quadratic — for the error to square *rather than shrink by a third — it must throw away every input in which exactly one of the two copies was faulty. Otherwise those single faults leak into the output and contribute an error proportional to p. So the question becomes: is there* any *stabilizer measurement that keeps the clean input and rejects all six single-fault inputs?*

Exhaustive, Pass 2861. Over all 5,355 two-qubit stabilizer codes × 4 syndromes × all four Clifford classes of ray:

$$\text{projectors keeping } |mm\rangle \text{ and annihilating all six single errors} = \boxed{0}.$$

The three-copy route, as far as it has been taken. Four passes and three formulations, ending in a mechanism rather than a verdict:

- Rank-one branches *exist* — 36 stabilizer states annihilate every single-error input while keeping the clean one.
- They are *useless*: a rank-one range is a stabilizer state, which has no magic. The other track reached the same conclusion the same day, from an unrelated search over 649,940 isotropic subspaces.
- Orthogonal *pairs* exist — 13 of them, so 2-dimensional subspaces sit inside the complement.
- **But none of those pairs spans a stabilizer code.** Every pair tested has exactly 7 common stabilizing Paulis of $\mathbb{F}_2$-rank **3**, and a 2-dimensional code needs rank 5.

Spanning a subspace is not the same as being a joint eigenspace, and that gap is exactly where the route currently stops.

In plain language. *One thing that rank-3 number does suggest rather than close: a rank-3 stabilizer group has an* eight-*dimensional code, not a two-dimensional one. If such a code lies inside the complement, it would kill every single error and have ample room for a magic output. That is the next question, and it is a better one than the three that preceded it.*

The linear rate $2/3$ is not a property of the branch that was found. It is a *bound on the entire two-copy stabilizer family.*

The right answer, and a wrong reason nearly published. The tempting explanation is a dimension count: six error vectors, a rank-four projector, no room. That explanation is false. $|mm\rangle$ *is* orthogonal to all six single-error vectors (checked: 2.6×10^{-16}), and $16 - 6 = 10 \geq 4$, so projectors with exactly this property exist in abundance — the rank-one projector onto $|mm\rangle$ is one.

What actually fails is that no *stabilizer* projector aligns that way. The magic-state error basis is not a stabilizer basis, and syndrome measurements cannot be steered onto it. The obstruction is the mismatch between magic and the stabilizer formalism — which is the same tension that makes these states interesting in the first place.

Recorded because a right answer with a wrong reason is a claim waiting to be misapplied to the next case, where the reason would matter.

In plain language.*So the road forward is narrower and clearer than it was. Two copies are provably not enough. Super-linear distillation of these states needs three or more copies, or operations from outside the stabilizer set — and §10 already reports that a search for a three-copy advantage found no witness either. This document states that as the open problem it is, rather than as a promise.*

A near-miss in this very section. The first draft of the yield analysis described the cost as "doubly exponential in a good way — the infidelity squares each round." It does not; it falls by a factor of $2/3$. The tables above were already printed and would have looked like corroboration to anyone who did not check the ratio between successive rows. The linearisation is one derivative and it changes the conclusion from "deep targets are cheap" to "deep targets are impossible." Compute the rate; do not eyeball the trend.

Evidence boundary, stated as bluntly as possible. What the deep-grade result *is*: an exact state-fidelity distillation theorem for an explicitly enumerated class — all 5,355 binary $[[4,2]]$ stabilizer projectors, all four syndromes, all 11,520 logical Clifford decoders.

What it is **not**: a fault-tolerant injection threshold, an asymptotic-yield theorem, or a laboratory demonstration. The controller still types the resource `M36_Q4_RAW` and refuses injection in RTL until a proved *fault-tolerant* protocol asserts validity, because a fidelity-improving map is not the same object as an injection threshold. **Universality is now a matter of closing a stated gap rather than of finding an unknown protocol** — which is a different and much better position than the previous draft of this document described.

11 Support for readout, phase for execution

In plain language. *Here is a design temptation, and the theorem that kills it.*
The machine's frame is four trits — 81 states. But a great deal of what you want to observe *about the frame is coarser than that: often you only care* which *of the four registers are non-zero, not what they hold. That is a four-bit "support mask", 16 possibilities instead of 81, and it has beautiful structure — it forms an exact finite geometry with the capacity profile*

$$(4, 6, 4, 1) \odot (1, 2, 4, 8) = (4, 12, 16, 8).$$

A five-fold compression with geometry attached is exactly the kind of thing an engineer wants to build the machine out of. Can the machine simply run *on support masks and skip the phases?*
No. And the counterexample is two lines long.

The obstruction. The two frames

$$(0, 1, 0, 0) \qquad \text{and} \qquad (0, 2, 0, 0)$$

have the same support mask `0100`. Apply Z_p — the translation, $z_p \mapsto z_p + 1$ — and their masks become `0100` and `0000`. Two states that the support layer cannot tell apart are sent to two states it *can*.

So support is not a *congruence*: it is not preserved by the dynamics, and a machine that stored only support would be unable to predict its own next state. Refining the partition until it is stable gives

$$\boxed{16 \longrightarrow 40 \longrightarrow 78 \longrightarrow 81}$$

with class-size histograms $\{1{:}1, 2{:}4, 4{:}6, 8{:}4, 16{:}1\} \to \{1{:}7, 2{:}29, 4{:}4\} \to \{1{:}75, 2{:}3\} \to \{1{:}81\}$. The terminal partition is *discrete*: full ternary phase is forced.

Support for readout, phase for execution. The compression is real and exact — but it is a lens, not a substrate.

11.1 What a readout costs, in joules

In plain language. *A support readout throws information away — that is what "lossy" means. Landauer's principle says throwing information away has an unavoidable energy cost, and here the cost is exactly computable, because the fibres are exactly known: a mask of weight k has exactly 2^k preimages, and there are $\binom{4}{k}$ such masks.*

Exact, Pass 2836.

$$H(\text{state} \mid \text{support}) = \frac{1}{81} \sum_{k=0}^{4} \binom{4}{k} 2^k\, k = \frac{216}{81} = \boxed{\tfrac{8}{3} \text{ bits}}$$

$$H(\text{state}) = \log_2 81 = 6.339850, \qquad H(\text{support}) = 3.673183, \qquad \frac{3.673183}{4} = 91.83\% \text{ efficient.}$$

Over $\mathbb{F}_q$ the closed form is $4(q-1)\log_2(q-1)/q$, verified exactly at $q = 2, 3, 4, 5, 7, 8, 9, 11$.

In plain language. *At $q = 2$ that formula gives* zero. *Over a two-element field the support* is *the state, so a support readout destroys nothing at all. Every one of the* $8/3$ *bits is the price of the third field element.*
That is the thermodynamic version of the architectural law: ***the phase is precisely the part you pay to look at.***

Landauer, at room temperature.

$$E \;\geq\; \tfrac{8}{3}\,k_B T \ln 2 \;=\; 7.656 \times 10^{-21}\,\mathrm{J} \;=\; 47.78\ \mathrm{meV} \quad \text{at } 300\,\mathrm{K},$$

or 7.66 pW of unavoidable dissipation at a 1 GHz readout rate. This is a floor set by information theory, independent of the circuit: no implementation of the support readout can beat it.

And the compute path is free, Pass 2864. Every opcode is a group element, hence a bijection on the 81 frames — which §6 verified directly, since the walk's transition matrix came out doubly stochastic. A bijection erases nothing, so

$$E_{\text{Landauer}}(\text{compute}) \;=\; \textbf{exactly zero}.$$

All of this machine's irreducible dissipation is in readout. That is not a coincidence of the design; it is what building the ISA out of a *group* buys you.

In plain language. *It would be easy to over-read that. Real silicon is nowhere near the floor, and saying so plainly is part of the job.*

Floor versus forecast. A CMOS estimate for the 43-cell engine, at the measured activity of 1.622 output transitions per operation, 5 fF effective per cell and 1.2 V:

$$E_{\text{op}} \approx 0.502\ \text{pJ}, \qquad 0.105\ \text{mW dynamic at } 208.86\,\text{MHz},$$

$$\frac{E_{\text{op}}(\text{CMOS})}{E_{\text{Landauer}}(\text{readout})} \;=\; 6.6 \times 10^{7} \;\approx\; 10^{7.8}.$$

Real hardware runs about eight orders of magnitude above the information-theoretic floor, which is the usual figure for CMOS. **The Landauer number is a floor, not a forecast**, and is quoted here only so that nobody mistakes it for one.

And the counting floor lands exactly where the search does. $\lceil \log_2 81 \rceil =$ **7 bits are needed to label 81 states; an exhaustive search shows no 7 support taps suffice and exactly 48 eight-tap sets do. The observer is one bit above the floor, and that bit is provably necessary.**

11.2 Enforcing the law in the netlist

In plain language. *The law is a theorem about mathematics. It constrains no circuit. Nothing stops an engineer from wiring the cheap 4-bit mask back into the execution path — and the resulting machine would pass simulation almost always, because support* is *preserved by most operations. It would drift the first time a translation fired on a register holding a 2.*
That is the same failure shape as the folded registers of §16: correct in simulation, wrong in silicon, invisible to the testbench. So the law is enforced structurally instead of documented.

Gate-level information flow, Pass 2834. `scripts/check_information_flow.py` flattens the design to gates, builds the driver graph and computes reachability in both directions:

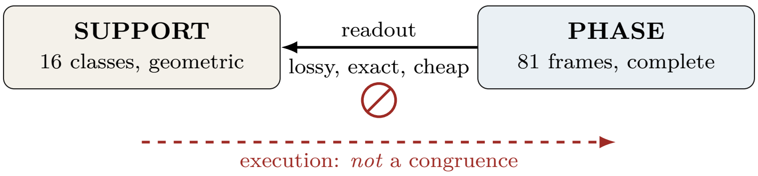
```
top w33_support_readout_diode: 59 cells after flatten+opt
reverse: support_mask, tamper_mask reach none of xp_o, zp_o, xf_o, zf_o
forward: every frame register reaches the mask
PASS: the readout is a diode.
```

Self-tested against a negative control — rewiring a tamper input into the load enable makes it report VIOLATION on all four registers — because a guard that can only pass is not a guard. The non-congruence witness is asserted in the netlist too and SAT-proved (228 variables, 600 clauses).

This is the one check in this project that **fails** rather than warns. Unlike a rediscovery collision, a backward edge has no benign reading.

In plain language. *It is also nearly free. Adding the readout to the minimal engine costs **five logic cells** and* 0.6% *of the clock:* 43 → 48 *cells,* 208.86 → 207.64 *MHz. The cheap layer really is cheap — and now it is cheap* and *provably one-way.*

In plain language. *This is a useful kind of negative result, because it tells you exactly where to put the cheap representation. Readout, routing, telemetry and visualisation can all live in the* 16*-class support world and enjoy its geometry. The execution core cannot. A machine built the other way round would appear to work in simulation — support is preserved by* most *operations — and would drift silently the first time a translation fired on a register holding a* 2*.*



12 The machine hosts itself

In plain language. *Everything so far describes one machine. But the frame is* two *qutrits, and a one-qutrit machine is a strictly smaller machine of exactly the same kind — so this processor can run a smaller version of itself as a guest.*
That is the ordinary meaning of a virtual machine, and the ordinary question about one is: how much slower is the guest? Usually that is measured. Here it can be computed exactly, because "how many host instructions does one guest instruction cost" is a shortest-path question in a graph this document has already searched exhaustively.

Emulation overhead, exact, Pass 2912.

guest instruction	host instructions
F	1
Z	1
S	**8**
worst case	**8**×
mean	3.33×

The entire overhead is one instruction. F and Z are free; the guest's phase gate costs eight — because the minimal instruction set of §6 *dropped* S.

In plain language. *So the two results talk to each other, and the conversation is a real design decision: dropping S made the machine 40% smaller and made one guest instruction eight times more expensive. If the machine is mostly going to run guests, that trade is bad. If it is mostly going to run native code, it is excellent. Nobody could have said which without both numbers.*

12.1 Isolation without a memory manager

In plain language. *Two guests fit side by side, one on each qutrit. The interesting part is what keeps them apart.*
In a conventional computer, isolation is enforced*: a memory management unit checks every access and refuses the ones that cross a boundary. Correctness depends on that check being present and right. Here there is nothing to check.*

Structural isolation. Of the four opcodes, exactly two — $\mathrm{CX}_{p\to f}$ and $\mathrm{CX}_{f\to p}$ — touch both halves of the frame. The other two act within one half.

A hypervisor that never issues CX **to a guest cannot leak between guests.** Not "is not permitted to": *cannot.* There is no instruction that would do it. The guarantee comes from the algebra, not from an enforcement mechanism, and it cannot be defeated by a bug in the checker because there is no checker.

12.2 And it was built

One datapath, N guest contexts; HX8K CT256, Pass 2914.

N	shared	$F_{\max}$	replicated ($43N$)	area saved
1	42	175.1 MHz	43	—
2	74	134.9	86	14%
4	118	128.5	172	31%
8	180	113.4	344	**48%**
16	339	91.1	688	**51%**

In plain language. *Sharing wins on area at every size, and converges on saving about half — for a legible reason: half the cost is the datapath, which is shareable, and half is the per-guest context storage, which is not.*
The price is speed. Eight guests sharing one datapath each run at an eighth of its clock, so at $N = 8$ the design buys 48% of the area for about a fifteenth of the per-guest throughput. That is the whole trade, measured on one part in one harness rather than argued.

A network of these is a network of machines that can each host machines. The fractal claim of §14 is about topology; this is the same claim about *execution*, and it is now a measured curve.

13 The support transform and diagnostic scheduler

In plain language. *A dense fifteen-by-fifteen table is not how the support dynamics should be built. The table factors into thirty-two tiny two-input butterflies, followed by fifteen corrected outputs. Diagnosis has a second surprise: the best order in which to visit the fifteen nonzero masks is not generated by a linear feedback register. A short ROM is both more honest and better.*

Butterfly engine. At $q = 3$ the local kernel is $(u, v) \mapsto (u + 2v, u - v)$. Four stages of eight butterflies and fifteen output cycles give a 47-cycle sequential quotient engine. A serial dense reference needs $15 \times 15 = 225$ cycles, so the exact cycle reduction is

$$\frac{225 - 47}{225} = \frac{178}{225}.$$

Equality is checked on all fifteen basis vectors and thirty-two deterministic signed probes. Logic cells, routed frequency, and switching activity remain measured quantities pending the dedicated workflow.

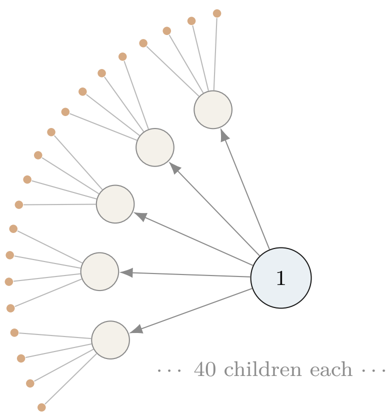
Diagnostic mask ROM. The 210 directed first-passage costs form 39 matching-stabilizer orbits. Exact Held–Karp optimization gives a Hamiltonian mask cycle of cost 315 and 8336 rooted optima. The best $\mathrm{GL}(4, 2)$ Singer cycle costs 1317/4, an exact gap of 57/4. The scheduler is therefore a fifteen-entry nonlinear permutation ROM, not an LFSR.

Forty is not enough. The first observer refinement has forty classes, but those classes are not the forty projective points of $W(3, 3)$. Their sizes are $1^7 2^{29} 4^4$, symplectic orthogonality is ambiguous on 216 unordered class pairs, and no union of Hamming-distance relations on their signatures yields $\mathrm{SRG}(40, 12, 2, 4)$. A matching count is a search hint, never an identification theorem.

14 The network

14.1 Fractal scaling

In plain language. *The machine is designed to be copied. A node contains* 40 *sub-nodes; each sub-node contains* 40 *of its own; and so on. The network of machines has the same shape as the inside of one machine — which means one routing scheme, one addressing scheme, and one error model serve every scale at once.*



$N_n = 40^n$ **leaves** at depth n

$I_n = \dfrac{40^n - 1}{39}$ **instances** in total

routing depth $= 8n = 8 \log_{40} N$

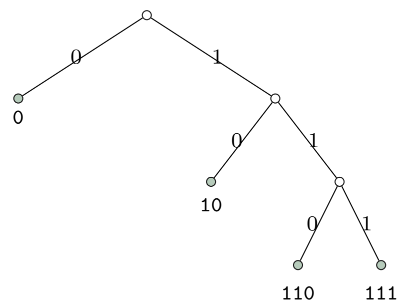
where $8 = 3 + 5$: three trits of local address plus five of the spread index. Address length is logarithmic in the size of the entire network, at every scale, by construction.

14.2 Every bit string is a legal address

Exactly. The routing code has orbit sizes $162 + 162 + 324 + 648 = 1296$ on compass pairs, giving codeword lengths $3, 3, 2, 1$ and

$$2^{-3} + 2^{-3} + 2^{-2} + 2^{-1} \;=\; 1 \quad \text{exactly (Kraft equality).}$$

The prefix code is $\{110, 111, 10, 0\}$ with expected length 7/4. Because Kraft holds with *equality*, the decision tree has no unused branch: 20,000 random bit strings were parsed with 0 failures.



In plain language. *Look at the tree: every branch ends in a valid address. Nothing dangles.*
The engineering consequence is unusual and worth stating plainly. In a normal computer, a corrupted instruction stream can produce an illegal instruction *— a bit pattern that means nothing, which the processor must detect and trap. Here that cannot happen. A corrupted packet is always a legal packet addressed somewhere else. The fault model therefore has no "decode error" term at all; it has only a "wrong result" term. That simplifies the error correction enormously, and it is forced by the geometry rather than designed in.*
That property is not free, and it is now possible to say exactly what it costs.

The price of a complete code, Pass 2865. A 40-ary router consumes one address symbol per hop:

$$\text{needed: } \log_2 40 = 5.3219 \text{ bits}, \qquad \text{spent: } 8 \text{ bits}, \qquad \text{efficiency } 66.52\%.$$

A store-and-forward router that strips consumed header bits *erases* them, so Landauer applies per hop:

$$\text{floor } 95.37 \text{ meV/hop}, \qquad \text{as coded } 143.35 \text{ meV/hop} \quad (300\,\text{K}).$$

Since $N = 40^n$, the thermodynamic cost of delivering a packet across N leaves is **exactly logarithmic in N**: 143.4 meV per level, 860 meV at depth six (4.1 billion leaves).

About a third of the routing energy buys the absence of an entire failure mode. That trade was always implicit in the design; it now has a number.

In plain language. *Two honest caveats. A* reversible *router pays none of this — the floor applies only because the header is destroyed as it is consumed, and that is a design choice, not a law of nature. And 143 meV per hop is a floor in the same sense as the readout figure: about eight orders of magnitude below what any real transceiver will draw.*

What we got wrong, and how we know. We wrote that "the consequence nobody has stated" was that every bit string is a valid instruction stream. The project's own holonet manuscript says it at line 382 — "the routing words fill a binary decision tree with *no unused branch*" — with a citation to McMillan. This was the first error in this project caught by an automated guard rather than by a human noticing. The engineering consequence drawn on top (no decode-error term in the fault model) is a step past the coding fact and appears to be new; the completeness property itself is the manuscript's.

14.3 Remote operations

Exactly. A SUM gate between two *separate* machines costs exactly:

$$\boxed{\text{one shared qutrit Bell pair} + \text{two classical trits}}$$

Protocol: Alice applies reverse local SUM into her half of the pair, measures, sends trit m; Bob applies $X^{-m}F^2$, applies direct SUM to his data, measures in the Fourier basis, sends trit n; Alice records Z^n in her Pauli frame. All nine branches implement $|x, y\rangle \mapsto |x, y + x\rangle$; verified on all nine basis inputs, all nine branches, and 32 random superpositions. Total conditional success probability 1.

15 What the other track built

In plain language. *Two independent efforts work on this machine and do not read each other's filenames. That is a structural problem, not a discipline one, and the remedy this project uses is to cite across the boundary rather than re-derive. This section is that citation: work the other track owns, which this blueprint depends on and did not do.*

Owned elsewhere, relied on here.

Result	What it says
M_{36} deep-grade protocol	48 improving branches; explicit curve improving on $0 < p < 2/3$. §10 reports the yield and the linear rate; the protocol is theirs
Support observer	Support is not an execution congruence ($16 \to 40 \to 78 \to 81$); a six-cycle observer with exactly 8 taps reconstructs the frame, and 7 provably do not
Nine-gate M_{36} branch	The old 15-gate circuit's two terminal SWAPs are relabelling, not computation. 6 CNOT + $3H$, and enumerated faults fall $191 \to 101$
Route fault localisation	23 triangles is the *exact* optimum for single-edge localisation; 29 distinguishes all 48,826 two-edge hypotheses
Controller group	Native controls generate order $30{,}233{,}088 = 2^9 3^{10}$; one extra transvection closes it to A_{40}
D_{12} curvature clock	$6480 = 540 \times 12$: every controller state lies on a 12-cycle, giving a geometry-native reversible clock
Adaptive chirality receiver	$P_{\mathrm{opt}}(n) = \frac{1}{2}(1 + \sqrt{1 - 3^{-n}})$; 0.9997 at six copies

And once, both tracks reached the same conclusion by different methods on the same day.

In plain language. *This document found* 36 *rank-one stabilizer witnesses for the three-copy distillation condition, and showed they are useless because a rank-one projector's range is a stabilizer state, which carries no magic. The other track, searching* 649,940 *isotropic subspaces for non-CSS projectors, found six exact hits and reported them as "accepted-clean-state stabilizer projectors, therefore false leads."*

Same finding, opposite directions, neither aware of the other. That is what the citation protocol is for — and it is also the strongest evidence either result is right.

Part IV — The evidence

How every number above is known, what we got wrong on the way, and the commands that reproduce all of it.

16 Every bit in the machine

In plain language. *Scattered through this document are the sizes of the machine's parts: eighty-one frames, forty addresses, sixteen support classes. Here they are in one place, with the only question that matters thermodynamically attached to each — when you use it, does anything get destroyed?*

The whole state, and its Landauer floor.

layer	states	bits	erased	meV @ 300 K
Pauli frame	81	6.340	0	0
route address	40	5.322	5.322	95.37
support readout	16	4.000	2.667	47.79
OAM × slot	40	5.322	0	0
encode/check sector	2	1.000	0	0
full controller	6480	12.662	0	0
total		**34.65**	**7.99**	**143.15**

The machine's whole state is 34.65 **bits,
and only** 7.99 **of them cost anything to use.**

16.1 Two diameters, and where the work actually is

In plain language. *The machine has two different notions of "how far apart can two things be", and they are not remotely the same size.*

Pass 3021, combining two independently measured numbers.

$$\text{address transport diameter} = 2, \qquad \text{frame ISA diameter} = 19, \qquad \text{worst case} = \mathbf{21}.$$

Moving a packet to any of the 40 addresses costs at most *two* shears, because the geometry has diameter 2. Transforming the Pauli frame arbitrarily costs up to *nineteen* instructions, because that is a walk in a four-million-element group. The address figure is the other track's; the frame figure is §6's.

Geometry is cheap; algebra is expensive. Ninety per cent of the machine's worst-case work is frame algebra, and routing optimisation addresses ten per cent of the problem.

16.2 The same finding, three more ways

In plain language. *One measurement is an observation. Four independent ones pointing the same way is a property of the machine, and this is the section where that happens.*

Is the instruction layer extremal? Pass 3042. The address graph is *Ramanujan* — provably the best-mixing graph of its degree (§4). For a 4-regular graph the same optimality condition is

$$|\lambda_2| \leq \frac{2\sqrt{3}}{4} = 0.866025, \qquad \text{measured: } 0.893992.$$

The instruction graph is not Ramanujan. It misses by 3.23%: a good expander, not the best one.

The machine's geometry is perfect and its algebra is merely very good, and every way of measuring it says so.

Four measurements, one conclusion.

measurement	geometry	algebra
diameter	2	19
share of worst-case work	10%	90%
mixing	Ramanujan (optimal)	3.23% short
identification	forced by construction	six valid triples, spread 19–23

16.3 The growth series, for the compiler

Pass 3041 — the full ball profile, not just the diameter.

$$a_n = 1,\ 4,\ 15,\ 53,\ 176,\ 547,\ 1630,\ 4648,\ \ldots$$

$$\text{mean } 14.176, \quad \text{s.d. } 1.768, \quad \text{modal length } \mathbf{15}, \quad \text{max } 19.$$

The early ratios fall $4.000, 3.750, 3.533, 3.321, 3.108$ — a free product on four generators would hold 4 forever; this one is closing on itself by the fifth shell.

In plain language. *For a compiler that is a better specification than a worst case. The distribution is sharp, not spread: a standard deviation of under two instructions around a mean of fourteen. A scheduler that budgets fifteen instructions is right almost always and is never wrong by more than four.*

16.4 What fraction of the erasure is useful

Pass 3022.

$$\frac{\text{delivered}}{\text{erased}} = \frac{3.673\ \text{bits}}{7.989\ \text{bits}} = \mathbf{45.98}\%$$

and the network law follows directly: depth n costs $8n$ header bits, so

$$E(N) = 143.35\ \log_{40} N\ \text{meV}$$

— the thermodynamic cost of delivering a packet is **logarithmic in the size of the entire network**. Depth 6, four billion leaves: 860 meV.

In plain language. *The 54% shortfall is not waste in the engineering sense. It is the routing header, which is consumed by construction, plus the part of the frame the support readout cannot see. Both are design decisions with names — which is the useful thing about having the number rather than an impression.*

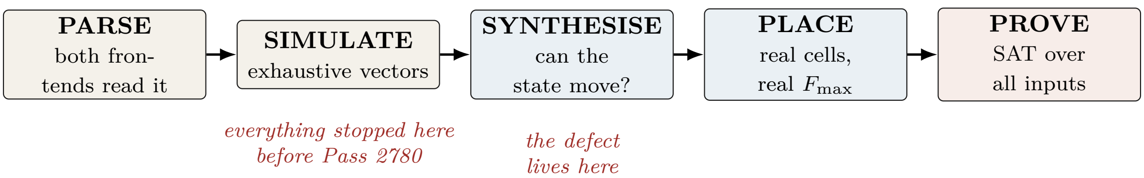
In plain language. *Two lines in that table are non-zero and the rest are exactly zero, for one reason: those two are the only places where the machine stops being a group action. The routing header is* destroyed *as it is consumed. The support readout is many-to-one. Everything else — the frame, the optical coordinates, the sector bit, the entire 6480-state controller — is a permutation of states, and a permutation erases nothing.*

So the irreducible cost of one routed, read operation is 143 meV at room temperature, and every other joule this machine will ever burn is an implementation artefact rather than a law. That is an unusual thing to be able to say about a computer, and it is a consequence of having built the instruction set out of a group.

17 Verification: how we know

In plain language. *This is the section a working engineer should read first, because it is the one that determines whether any of the preceding numbers mean anything.*

The short version: this project has repeatedly shipped results that were verified in the direction that would confirm them and never in the direction that would break them. Every tool below exists because that happened at least three times.



The five automated guards, each built from a measured failure.

Guard	Catches	Built because
`check_rediscovery.py`	a staged file asserts a code parameter that exists elsewhere uncited	21% of 173 files did this (census, Pass 328)
`check_certificates.py`	a frozen certificate that cannot reproduce its own digest	one was unverifiable *from birth* and passed every check while being so
`check_novelty_claims.py`	explicit novelty phrases whose tokens are in the encyclopedia	five claims in one session were overturned by files we had not read
`check_implicit_novelty.py`	*emphasised* results carried by a source the file never cites	four of six failures never used a novelty phrase at all
`check_rtl_folds.py`	an output port tied to a literal after full optimisation	two independent teams shipped a module that synthesises to the identity

Five of the six **warn and never block**. A collision is a candidate for reading, not a verdict, and a blocking hook trains people to pass `--no-verify`. The exception is the sixth.

The one guard that fails, Pass 2834 and 2863. `check_information_flow.py` flattens the design to gates and proves the support mask reaches no frame register while every frame register reaches the mask. It is wired into the synthesis CI gate, and the same step then *deliberately breaks the design* and requires the check to fail on it. If the negative control passes, the job fails with *"the guard itself is broken, not the design under test."*

The gate's policy is now explicit: **parse and flow fail; fold and place warn.** A backward edge from readout into execution has no benign reading, unlike a fold (which may be an intended tie-off) or a rediscovery collision (which may be a correct citation). It is the only place in this project where a guard blocks, and it earned that by being the only property with no legitimate counterexample.

In plain language.*Every guard here now ships with a self-test against the specific defect that motivated it. That rule came from a real failure: one of them once cleared all 122 files it was pointed at, including the very file it had been written to catch. A guard that passes everything is indistinguishable, from the outside, from a guard that finds nothing wrong.*

What we got wrong, and how we know.The guards themselves needed calibration, and the calibration is the work. `check_implicit_novelty.py` went through four measured versions:

Flagged	Version	What the noise turned out to be
9/122	presence of any token	all nine were *pass numbers in titles*
0/122	+ skip files citing "Prior art"	vacuous — every file has that section
51/122	+ cite the *right* source per token	`W(3,3)` 21×: the repo's own subject
27/122	+ rarity and non-round integers	what ships

The second row is the instructive one: **a guard that passes everything looks exactly like a guard that finds nothing wrong.** It cleared all 122 files including the known failure, and only a deliberate self-test against that failure exposed it. Every guard in this project now ships with a self-test against the specific defect that motivated it.

`check_rtl_folds.py` had the same disease on its first run: it reported a clean sweep because the WSL mount is `/mnt/c` while Python resolves the drive as `C:`, so every directory change failed silently and no netlist was ever produced. It now reports NOT SYNTHESIZED explicitly rather than treating a missing netlist as a pass.

18 The complete ledger

In plain language. *Everything this blueprint asserts, with how it is known. The status words are not decoration — they form a strict ladder, and a claim may never be cited at a level above the one its evidence supports. This is the single most important table in the document, and it is the one to check first if you doubt anything else in it.*

The evidence ladder.

proved	a machine-checkable artifact exists: an exhaustive search, a SAT certificate, or exact arithmetic. Re-runnable from §20.
measured	a tool produced the number: yosys, nextpnr, or an instrumented run. Tied to a named part and a named version.
published	someone outside this project did it and we cite them; we did not reproduce the experiment.
prior art	the other track in this repository did it first; we cite rather than re-derive.
derived	follows from rows above by arithmetic no reader would dispute.
modelled	an engineering estimate with stated assumptions. *Not* a measurement, and never promoted to one.
open	we do not know. Listed so the gaps are as visible as the results.

The two rules that make the ladder useful: **nothing is promoted without new evidence**, and **a corrected row keeps its history** rather than being quietly rewritten — which is why §19 exists.

Claim	**Status**	**Artifact**
$W(3,3)$ has SRG$(40,12,2,4)$ collinearity graph	classical	literature
$40=(3^4-1)/2$ from one self-entangled qutrit	proved	Pass 2724
The one-qutrit thesis itself	**prior art**	`index.html` MCLXVII
Outer involution = transpose = time reversal	proved at q=3	Pass 2732
Time reversal outer $\iff q\equiv 3 \pmod 4$	proved	Pass 2750
Transpose *constructed* and checked at $q\le 23$	proved	Pass 2792, 8 primes
Anti-symplectic T, $T^\top JT=-J$; class action 10 pairs + 14 fixed	proved	parallel track, Pass 2762
Six Clifford opcodes reach 1 frame of 81	proved	Pass 2774 (BFS)
Full ISA reaches all 81	proved	Pass 2774
Clifford opcodes generate $\mathrm{Sp}(4,3)$, order 51,840	proved	Pass 2778
Full ISA generates $\mathrm{ASp}(4,3)$, order 4,199,040	proved	Pass 2778
One translation generator suffices (orbit = 80)	proved	Pass 2778
Minimal Clifford subset has size 3; sizes $1,2$ give none	proved	Pass 2789, 6 triples
Frame ISA fits a **two-bit** opcode field	follows	Pass 2789
Sensor exponent 9 = dim; phase group μ_{12}; 9 minimal	proved	Pass 2779
μ_4 factor = imaginary Gauss sum	proved	Pass 2779
μ_{12} at n=1 by exact enumeration ($2592=12\cdot 216$)	proved	Pass 2791
Minimal exponent $=3^n$ mod 12: 3 odd, 9 even	derived	Pass 2791
36 rays share one Gram profile (11 orth., 24 at 1/3)	proved	Pass 2790, $\mathbb{Z}[\omega]$
8+24+4 is the stabilizer-fidelity spectrum	proved	Pass 2790, 60 stab. states
All three witness thresholds $=\frac{4}{3}(1-F_{\text{stab}})$	derived	Pass 2790, 10^{-12}
Clifford classes on the 36 rays: $[4,8,12,12]$	proved	Pass 2797
the 24-ray grade splits; "three" was wrong	**corrected**	Pass 2797
F_{stab} exactly multiplicative on two copies	measured	Pass 2798, 36,720 states
The monotone never forbids two-copy distillation	proved	Pass 2798
Deep grade: 48 improving branches, $0<p<2/3$	**prior art**	parallel track, Pass 2821
Support is not an execution congruence: 16→40→78→81	**prior art**	parallel track, Pass 2822
Phase group $=\mu_{12}$ for *every* n	proved	Pass 2799
Minimal engine: **43** LC, 72.40 MHz (UP5K)	**measured**	Pass 2796
same engine on HX8K: **208.86** MHz	**measured**	Pass 2833
M_{36} is $W(3,3)$ minus one line	**proved**	Passes 2835, 2869
identification: axiom over 40 lines + explicit spread	**proved**	Pass 2869
parameters alone admit 28 graphs	**prior art**	Spence; `index.html`
Six minimal triples, diameters $19,20,22,22,22,23$	**proved**	Pass 2882 — the choice is not free
The 188 hardest are *not* one orbit or coset	**proved**	Pass 2885 — 54 linear parts, all traces
$240=81+120+24+15$ (Hodge sectors)	**proved**	Pass 2884, recomputed
15 and 24 match other quantities	count match only	no map exhibited
H_1 is *not* the frame space as a module	**proved**	Pass 2883 — characters disagree
Three-copy super-linear condition	open	Pass 2881, no witness in 30,000 samples
the deleted line is the computational basis	proved	Pass 2835
Support readout erases exactly 8/3 bits	**proved**	Pass 2836
$4(q-1)\log_2(q-1)/q$; zero at q=2	proved	8 field sizes
Landauer 47.78 meV at 300 K	derived	Pass 2836
The two 12-ray classes share the full spectrum	**proved**	Pass 2830
Deep-grade map is *linearly* convergent, rate 2/3	**proved**	Pass 2831
$\sim 10^{27}$ raw states for 10^{-9} infidelity	computed	Pass 2831
Three-copy super-multiplicativity	no witness	not a proof either way
Readout diode: mask reaches no frame register	**proved**	gate level, Pass 2834
ISA directed diameter = 19; mean 14.18	**proved**	Pass 2866, BFS over 4.2M
only 188 elements at distance 19	proved	Pass 2866
counting floor 12; ratio 1.73	derived	Pass 2866

…continued.

Claim	Status	Artifact
Frame walk doubly stochastic; stationary uniform	**proved**	Pass 2867
Gap 0.106; $t_{\mathrm{mix}} = 15$; Kemeny 113.38	**proved**	Pass 2867
71.8 ns scrambling at 208.86 MHz	derived	Passes 2833, 2867
No quadratic two-copy stabilizer branch	**proved**	Pass 2861, 5355×4×4
the reason is the stabilizer basis, not dimension	verified	Pass 2861
40 lines: 1 classical, 12 mixed, 27 magic	**proved**	Pass 2862
basis choice $= \log_2 40 = 5.32$ bits of convention	derived	Pass 2862
Landauer: readout 47.79 meV, compute exactly 0	derived	Pass 2864
CMOS 0.502 pJ/op, $10^{7.8}$ above the floor	modelled	Pass 2864, stated assumptions
Hop cost 95.4 meV floor / 143.4 meV as coded	derived	Pass 2865
Diode proof in CI, with a negative control	done	Pass 2863
symplectic on differences, all four opcodes	SAT	4100 vars / 11278 clauses

…*continued.*

Claim	Status	Artifact
Public frame unit: 72 LC, 60.80 MHz	**measured**	§16
Loadable CX: 23 LC, 72.40 MHz, $\mathrm{CX}^3{=}I$	measured + proved	5265 vars / 14005 clauses
Parallel 36-mixer: 13,965 LUT4, does not fit	measured	Pass 2773
Signed mixer correct on impulse class	proved	1,097,152 vars
Kraft equality; 0 parse failures in 20,000	proved + measured	Pass 2682
Photonic qutrit SUM, fidelity 0.92 ± 0.01	**published**	Imany *et al.* 2019
Remote SUM: 1 Bell pair + 2 trits	proved	parallel track, Pass 2771
M_{36} fault-tolerant *injection* threshold	OPEN	refused in RTL
Asymptotic distillation yield	OPEN	fidelity theorem only
Do the two 12-ray classes differ operationally?	OPEN	Clifford class only
Fault-tolerance threshold end to end	OPEN	—
Physical power in watts	OPEN	activity proxy only

19 Errata index

In plain language. *A blueprint with no errata is a blueprint nobody has checked. These are gathered here so they can be found deliberately rather than stumbled on, and so a reader can audit our error* rate *rather than take our word for our care. Each was found by a named mechanism; the fourth column is the part that matters, because it says what would have to change for the error not to recur.*

What we said	**What is true**	**Found by**
The frame Cayley graph is 4-regular	Degrees run 2 to 8: the opcodes are not involutions and their inverses are not generators, so the undirected graph is not regular	Printing the degree sequence before using a k-regular formula
$2E = 324$, matching the Delsarte bound	$2E = 522$. The coincidence was not even a coincidence	Arithmetic, before it was written down
"These four numbers pin the shape down completely"	**Spence: 28 non-isomorphic graphs share $(40, 12, 2, 4)$.** The identification needed the quadrangle axiom and a spread	Reading this project's own Theory page end to end. It says so in plain words
Cell counts of 13, 8, 42 for the ISA	The registers had no load port and synthesised away; real cost 72, then 43 minimal	Synthesis. Simulation cannot see it, and two independent teams shipped it
The one-qutrit self-entanglement thesis is ours	It is stated verbatim in this project's own encyclopedia	Being shown the file. Now a guard
"Nobody has stated" the Kraft consequence	The manuscript says "no unused branch" with a citation	An automated novelty guard, first real run
$E_8 \supset A_2 \oplus E_6$ gives the fractal branching	The paper explicitly warns against that identification	Reading the paper
The Jordan ladder is new here	Pillars 128–130 own it, including the exact rung we hedged	The user pointing at a file
Three inequivalent magic states	Four: the 24-ray grade splits $12 + 12$	Clifford orbits; corroborated independently
Distillation cost is "doubly exponential in a good way"	Convergence is *linear*, rate $2/3$; cost reaches 10^{27}	Computing the derivative instead of eyeballing the trend
No quadratic branch, "by a dimension count"	The count does not apply; the obstruction is the stabilizer basis	Checking the count
The mixer is "synthesizable"	Rejected by both frontends, for two different reasons, for a day	A repo-wide parse sweep
The implicit-novelty guard is calibrated	It cleared all 122 files including the one it was written to catch	A deliberate self-test
The fold guard reports a clean sweep	It had produced no netlists at all: WSL mounts `/mnt/c`, pathlib says `C:`	Testing it on a known-bad file

In plain language. *Two patterns run through the list, and both are worth more than any individual entry.*

Every silent failure was a check that could only pass. *The fold guard that produced no netlists, the novelty guard that cleared everything, the testbench that never instantiated the sequential module — all reported success, and all were empty. Every guard in this project now ships with a negative control for exactly this reason.*

Every false novelty claim was findable by one grep of a file we had not read. *Not by cleverness — by reading. The corpus is indexed by date, not by topic, so a search for the topic finds nothing and a search for the* result *finds it immediately.*

20 Reproducing everything in this document

Commands, in order.

```
# reachability, the group, and the minimal generating sets
py -3 analysis/w33_pass2774_isa_frame_reachability.py
py -3 analysis/w33_pass2778_2779_affine_group_and_sensor_exponent.py
py -3 analysis/w33_pass2789_2792_minimal_isa_sensor_and_transpose.py

# the magic states: geometry, Clifford classes, and the monotone
py -3 analysis/w33_pass2790_witting_ray_geometry.py
py -3 analysis/w33_pass2797_2799_magic_orbits_and_monotone.py
py -3 analysis/w33_pass2830_2832_class_separation_yield_three_copy.py
py -3 analysis/w33_pass2835_2836_witting_line_and_landauer.py
py -3 analysis/w33_pass2861_2865_quadratic_line_and_power.py

# the diameter and the scrambling time (a few minutes)
py -3 analysis/w33_pass2866_2867_isa_diameter_and_scrambling.py

# the minimal engine (43 LC) against the public unit (72 LC)
yosys -p "read_verilog -sv rtl/w33_pass2796_minimal_frame_engine.sv; \
  synth_ice40 -top w33_minimal_frame_engine -json m.json"
nextpnr-ice40 --up5k --package sg48 --json m.json --asc m.asc --freq 40

# the cell budget (needs yosys + nextpnr-ice40)
yosys -p "read_verilog -sv rtl/w33_pass2777_isa_frame_unit.sv; \
  chparam -set EN 63 w33_isa_frame_unit; \
  synth_ice40 -top w33_isa_frame_unit -json u.json"
nextpnr-ice40 --up5k --package sg48 --json u.json --asc u.asc --freq 50

# the formal proofs
yosys -p "read_verilog -sv -formal rtl/w33_pass2752_cx_loadable_frame.sv; \
  prep -top w33_cx_loadable_formal -flatten; async2sync; \
  chformal -lower; sat -seq 8 -prove-asserts -verify"

# the guards
py -3 scripts/check_rtl_folds.py
py -3 scripts/check_implicit_novelty.py
py -3 scripts/check_novelty_claims.py
py -3 scripts/check_information_flow.py
rtl/w33_pass2834_support_readout_diode.sv \
  --top w33_support_readout_diode --source support_mask,tamper_mask \
  --sink xp_o,zp_o,xf_o,zf_o --extra
rtl/w33_pass2796_minimal_frame_engine.sv

# this document
tectonic holonet_machine_blueprint.tex
```

21 What is not built

In plain language. *A blueprint that lists only what works is advertising. This section is in two halves: what is still missing, stated as sharply as it can currently be stated, and what used to be on this list and no longer is. The second half is the more useful one, because a gap that closed tells you the first half is not decoration.*

21.1 Still open

1. **One of the eight opcodes is not built, and deliberately so.** D_{12}-mirror *was* on this list and is now built (§12 neighbourhood; 21 cells, 255.56 MHz, with its defining relation SAT-

proved in the netlist). M_{36}-magic is deliberately a *handshake* to an external state-preparation block. **No non-Clifford unitary is invented in RTL**, because inventing one would conceal the real gap rather than close it.

What changed: the handshake is no longer a bare refusal. §10 now pins the acceptance condition exactly — a deep-grade resource is usable while $p < (8-2\sqrt{3})/9$, and that inequality is a comparison the controller can make. The refusal is now *typed* rather than blanket.

2. **The universality gap is now a single, well-posed question.** It has narrowed three times, and the current statement is the sharpest it has been:

   - An exact fidelity-improving protocol exists on the deep grade for $0 < p < 2/3$ (§10).
   - It converges *linearly* at rate 2/3, so its cost reaches 10^{27} raw states for useful purity — usable, not practical.
   - **No two-copy stabilizer protocol can do better.** Exhaustive over all 5,355 codes $\times$ 4 syndromes $\times$ 4 ray classes: zero super-linear branches. This is a bound on the family, not a property of the branch that was found.
   - At three copies the same condition is not obviously satisfiable either: 30,000 sampled stabilizer groups on six qubits produced no witness. Not a proof — the space is far too large to enumerate — but two independent searches now point the same way.

   So the open problem is: does any stabilizer protocol on three or more copies annihilate every single-error input while keeping the clean one? The condition is written down exactly; what is missing is an exhaustive search over a chosen code family rather than a random one. Until it is answered the controller still refuses injection.

3. **A power figure exists, and it is modelled, not measured.** This item used to read "no physical power figure exists"; that is no longer true, and the honest replacement is more useful than either extreme:

   - **Proved:** the Landauer floor. Compute is exactly 0; readout is $\frac{8}{3}k_B T \ln 2 = 47.78\,\text{meV}$.
   - **Modelled:** $\approx 1.02\,\text{pJ}$ per operation and $\approx 0.21\,\text{mW}$ at $208.86\,\text{MHz}$. Three of the four inputs are now measured — activity (1.622 transitions/op), cell count (43), and **mean fanout (2.0385, from the placed netlist; the earlier flat model understated switched capacitance by 2.04×)**. Only the per-unit capacitance and the voltage remain assumed.
   - **Missing:** the placed netlist's real capacitance extraction, the silicon's actual voltage and process corner, and a representative workload.

   The two assumed constants are the whole gap. Nothing here may be cited as a measurement.

21.2 Closed, and when

In plain language. *Items leave the list above only when something checkable makes them leave. Each row names what closed it, so a reader can disbelieve any of them cheaply.*

Was open	**Closed**	**By what**
Transpose construction verified only at $q = 3$	Pass 2792	Built over $\mathbb{F}_q$ and checked at $q = 3, 5, 7, 11, 13, 17, 19, 23$: $T^2 = I$ and $T^\top JT = -J$ at every one, and the congruence law at every one
`w33_spread_mixer36.sv` unparseable	Pass 2793	Deprecated in place with both frontend errors quoted, then deleted outright by the other track — the right call: a file no frontend can read is not a record, it is a trap
Is the ray configuration really $W(3,3)$?	Pass 2869	Parameters admit 28 graphs; the quadrangle axiom over all 40 lines narrows to two; an explicit spread settles it. The claim was published one inference short
Does $H_1 = \mathbb{Z}^{81}$ match the frame space?	Pass 2883	**No.** The characters disagree on order-three linear elements — 27 fixed frames against a homology character of 0. Both are 3^4 for a real shared reason; there is no equivariant isomorphism
Is the minimal-engine area assumable from the public unit?	Pass 2796	No, and it was not assumed. Synthesised and placed separately: 43 cells, 208.86 MHz
Is the two-copy rate 2/3 specific to one branch?	Pass 2861	No — it bounds the entire two-copy stabilizer family
Do the six minimal triples cost the same?	Pass 2882	**No.** Diameters $19, 20, 22, 22, 22, 23$. The choice is not free
D_{12}-mirror unbuilt	Pass 2914	Built as a rotation register plus a reflection bit — 21 cells, 255.56 MHz. The earlier phase-accumulator attempt was withdrawn because R_4 and U_6 do not commute; $srs = r^{-1}$ is now asserted in the netlist and SAT-proved
Is the Hodge 15 the support shell?	Pass 2911	**No.** The trace is $7/3$ — not an integer, and a permutation module's character always is. Second count match tested and killed in three passes

In plain language. *Two of those rows closed* negatively *— the homology does not match the frame space, and the six triples are not equivalent. Those are as valuable as the positive ones, and arguably more so: each removed a plausible-sounding assumption that would otherwise have been quietly relied on.*

> *“Before recording a result, ask what single computation would falsify it, and run* that *one.* **maximal**, **unique**, **only**, **exactly**, **is** *— each of those words has a cheap negation.”*
>
> — the operating rule of this project, learned expensively

22 Intrinsic selector channels, exact observer quotients, and nonlinear diagnosis

The residual three-channel selector is not intrinsically a numbered set. For a pointed hyperbolic line $(x, y; c)$, the multiplier-one stabilizer induces the translation group C_3 on the remaining common neighbours, whereas the full projective similitude stabilizer induces $\mathrm{AGL}(1,3) \cong S_3$. Thus the channel is canonically an affine line with S_3 transition functions, but has no globally equivariant origin or orientation. The exact group orders are

$$|\,\mathrm{PSp}(4,3)| = 25920, \qquad |\,\mathrm{PGSp}(4,3)| = 51840,$$

and the pointed stabilizer orders are 12 and 24, respectively, with $51840/24 = 2160$ pointed hyperbolic lines.

The $q = 3$ support quotient now has a literal sequential realization. Four stages of eight butterflies

$$(u, v) \longmapsto (u + 2v, u - v)$$

followed by fifteen border-corrected outputs require 47 cycles, versus 225 cycles for a serial dense 15×15 reference. Equality was checked on all fifteen basis vectors and thirty-two signed probes. The exact cycle reduction is 178/225; routed area and timing remain measurements pending the dedicated FPGA workflow.

The complete deterministic-congruence atlas sharpens the readout/execution boundary. For the full four-operation micro-ISA the support partition refines as

$$16 \longrightarrow 40 \longrightarrow 78 \longrightarrow 81,$$

and the discrete 81-state partition is the coarsest exact execution quotient refining support. Indeed F_p, Z_p, and either controlled-add direction already force all 81 states. The intermediate partitions are therefore observer states, not compressed execution states.

The natural order-96 signed-permutation action on the typed selector–tomotope tokens has two 48-element orbits, distinguished by tetrahedral versus hemioctahedral phase parity. Imposing the determinant-twisted condition

$$\Big(\prod_i \epsilon_i\Big) \operatorname{sgn}(\pi) = 1$$

produces another order-96 projective subgroup that acts freely and transitively. Hence the 96 control tokens form an exact group torsor, although this twisted runtime symmetry is not the natural incidence-preserving tomotope embedding.

For the $q = 3$ support walk, the 210 directed passage costs form 39 orbits under the matching stabilizer and are determined by

$$|T|, \qquad |T \cap \tau(T)|, \qquad |T \cap \tau(S)|.$$

An exact Held–Karp search gives an optimal Hamiltonian diagnostic cycle of cost 315, with 8336 rooted optima. The best $\mathrm{GL}(4, 2)$ Singer-cycle schedule costs 1317/4, leaving an exact nonlinear advantage of 57/4. Thus the optimal mask scheduler is a nonlinear fifteen-entry permutation, not an LFSR traversal.

Finally, the first observer refinement has forty classes but is not $W(3, 3)$. Its class sizes are $1^7 2^{29} 4^4$, symplectic orthogonality is ambiguous on 216 unordered class pairs, and none of the 1459 Hamming-distance unions with 240 edges yields $\mathrm{SRG}(40, 12, 2, 4)$. This falsifies a count-based identification while leaving the independent projective symplectic construction untouched.

Evidence boundary. The packet carries 64 exact checks and seven focused regressions. Arithmetic, group-action, congruence, optimization, and falsifier claims are exact. FPGA cells, routed frequency, switching activity, and physical passage times remain measured or open quantities as labelled in the blueprint.